Commissioning Manual

Trimble® Earthworks Grader

For SITECH Only

Version 2.24.x
Revision A
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Contents

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Grader commissioning checklists

In this chapter:

1.1 Checklist—Pre-installation

Use the following checks to identify problems with a machine before an installation. Problems that are identified by this procedure must be corrected before commissioning is begun. Poor machine condition can affect the performance of the automatic blade guidance.

- ☐ Operate the machine and observe the performance of the hydraulic system, particularly the responsiveness of the hydraulic system to control movements. Any hydraulic system issues will remain after installation, and may be worsened if an Autos system is installed.
- ☐ Check the cutting edge:
  - Even vertical wear over the entire length of the blade. 0 ± 5 mm horizontal deviation.
  - For more information, see Chapter 3.2, Check machine wear.
- ☐ Check the circle play:
  - Follow the manufacturer's instructions and recommended tolerances.
  - For more information, see 3.2.4 Check the circle play.
- ☐ Check the circle backlash, if applicable to the attachment:
  - 0 ± 12 mm measured on the outside of the circle.
  - No play in connection between hydraswivel and hose tray.
  - No horizontal play from hose tray slide rod guides.
  - For more information, see 3.2.5 Check the circle backlash.
- ☐ Check the lift cylinder backlash:
  - 0 ± 3 mm difference between blade and circle when blade is lifted off ground.
  - For more information, see 3.2.6 Check the lift cylinder backlash.
- ☐ Check the lift cylinder blade joints:
  - 0 ± 1 mm difference between cylinder rod and the top of the circle, with blade hanging from the circle.
  - For more information, see 3.2.7 Check the lift cylinder ball joints.
- ☐ Check the blade rail shims:
  - 0 ± 1 mm clearance between blade and wear strips.
  - For more information, see 3.2.8 Check the blade rail shims.
- ☐ Check the tire wear and pressure:
  - Tandem tires are the same make, model, and size, and with approximately even wear and inflation.
  - Front tires the same make, model, and size, and with approximately even wear and inflation.
  - For more information, see Chapter 3.2, Check machine wear.

- ☐ Machine ECM firmware:
  - Contact the machine manufacturer to ensure that the ECM firmware is up to date.
- ☐ On electro-hydraulic (EH) machines perform a:
  - Joystick calibration
  - Solenoid or cracking current calibration. For more information, see 3.3.7 Perform a solenoid calibration.

1.2 Checklist—Configuring a system

- ☐ Connect all system components:
  - EC520
  - Display
  - GNSS receivers
  - MT900 target (optional)
  - ST400 sonic tracer (optional)
  - LR410 laser receiver (optional)
  - EM410 mast (optional)
  - Internet gateway (optional)
  - Radio (optional)
  - The GS5xx grade slope sensors appropriate to your system
  - RS400 rotation sensor
- ☐ Turn on the ignition to start the display and to provide power to the EC520.
- ☐ Confirm that the display's date and time are correct. To change the date and time, open Settings > System > Date & Time > Time Zone and select the correct time zone.

  Complete the EC520 firmware installation as per the release notes.
- ☐ Tap the Web Interface icon and log in.
- ☐ If this is the first Web Interface log in:
  - set an admin password
  - configure Wi-Fi
  - complete the setup section of the Install Assistant

  Otherwise, if this is a subsequent Web Interface login, the home page opens.
- ☐ Open Advanced > Licenses and apply a core bundle license. You can also apply any module licenses for the EC520.
- ☐ Open Configure > Install Assistant and complete each section. Use a TD5x0 or Wi-Fi BYOD display to perform valve calibrations.
- ☐ If using GNSS for 3D guidance, select Network > GNSS Correction Source and select a correction source.
- ☐ If using GNSS for 3D guidance, select Monitor > GNSS Details and check the GNSS receiver.

- ☐ Close the Web Interface, open the Operator App, log in and log out again. If the Operator App launcher is not installed, use MC Installer to install it.
- ☐ Open the Web Interface's Advanced > Licenses screen and apply the remaining module licenses to the display.
- ☐ If you have a WorksManager/Connected Community subscription:
  - Select Network > GNSS Connected Community; set up WorksManager/Connected Community.
  - Upload project files to WorksManager/Connected Community to sync to the machine.
  - Select Operation > Projects and set up the interval the EC520 synchronizes with WorksManager/Connected Community.
- ☐ Close the Web Interface again, and re-open the Operator App.
- ☐ If you are using a USB flash drive to transfer project files onto the system, insert and transfer from the USB flash drive to the machine.
- ☐ Confirm that all tiles on the Dashboard are not red. The Start button is then enabled, and tapping the button opens the work screen.
- ☐ Touch and hold on the text ribbon to configure the text items.
- ☐ Confirm that guidance is correct:
  - 2D – confirm system accuracy using the steps in the machine-specific *Measure-up and Sensor Calibration Validation Guide*.
  - 3D – confirm the Northing, Easting and Elevation position at the Focus tip.

  Test the system gives accurate position while working on a design.

2

Configuration

In this chapter:

2.1 Install the firmware onto EC520

> **NOTE –** The following procedures assume the use of a TD5x0 display as the upgrade device. If significantly different steps are required for a PC, the differences are shown as a Tip.

2.1.1 Gather required firmware

Download the latest .sg6 firmware from Partners using:

- USB flash drive – copy the firmware onto a USB flash drive. Insert the USB flash drive into the TD5x0 display's USB port.
- PC – copy the firmware onto the PC.
- TCC – copy the firmware to the machine-specific Trimble Connected Community EarthworksData folder. Synchronize the EC520 with TCC.

Apply the firmware in the Web Interface on Monitor > Onboard Devices. This process takes more than five minutes to complete.

NOTE – This firmware upgrade is for 2.24.x. Once the firmware is upgraded to 2.24.x, the system will limit how far you can downgrade the firmware in future.

NOTE –If using WorksManager to remotely apply firmware, we recommend that you copy the firmware into the machine-specific EarthworksData folder. Avoid using the Remote EC520 portal.

NOTE –When you use an SG6 file to update the software on your EC520, it replaces the files ww15mgh.ggf and xintcoordsys.xml. If you are using a custom geoid file named ww15mgh.ggf, the file will be replaced which may affect guidance.

2.1.2 Obtain the required licenses

Contact your SITECH dealer to purchase the licenses required for the installed system:

- Using a TD5x0 display—copy the licenses onto a USB flash drive. Insert the USB flash drive into the display's USB port and turn on the display.
- Using a PC—copy the licenses onto the PC.
- Using Cloud Licensing—In the Web Interface, turn on Auto Sync or use the Sync Now button on the Licenses page.

At a minimum, before you can install new EC520 system ECM firmware, ***you must have***:

- A Core License installed.
- The Core license must provide a software maintenance date ***ON*** or ***AFTER*** March 2026. To determine the software maintenance date, refer to Software Maintenance Until item in the Web Interface > Advanced > Licenses screen.
- Completed the First Time Setup.

For information on installing licenses, refer to Chapter 2.8, Set up licenses for this machine.

TIP – The firmware installation requires a Wi-Fi or physical connection to the ECM. If the ECM's Wi-Fi access point is not enabled:

- Use the TD5x0 display—connect the display to the harness in the usual way.
- Use a PC—use a standard Ethernet cable and an Ethernet Service Adapter (P/N150424-01) to connect the PC to the ECM.

2.1.3 Connect to the EC520

The firmware installation requires a physical connection to the EC520:

- Using the TD5x0 display – connect the display to the harness in the usual way.
- Using a PC – connect the PC to the EC520 using a standard Ethernet cable and the Ethernet Service Adapter (P/N 150424-01).

2.1.4 Power-up the EC520 controller

Connect the controller to a reliable power source.

2.1.5 Connect to the EC520 controller via the Web Interface

1. Connect the upgrade device (either a TD5x0 display or a PC) to the EC520 controller using the required cables.
2. Select the Web Interface icon on the TD5x0 display. The install screen appears.

> **TIP –** Browse to the controller Web Interface IP address: 192.168.168.1. The install screen appears.

3. Select Choose File:

   If using a PC – browse to the downloaded firmware file on the PC's file system, and select the file.

   If using the TD5x0 display:

   a. Select Documents. The Recent screen appears.
   b. Select the menu icon in the top right corner of the screen.
   c. Select Show SD card.
   d. The USB flash memory appears as a folder in the left hand menu. Browse to the new firmware file and select the file.
4. Select Install OS. Firmware upload begins.
5. When the upload completes the firmware installation begins, and a progress screen appears.
6. Once the installation completes and the controller finishes rebooting, select the Web Interface icon. The Web Interface appears.

> **TIP –** Once the installation completes and the controller finishes rebooting, browse to the controller Web Interface IP address: 192.168.168.1. The Web Interface appears.

2.2 Recommended TD5x0 firmware versions

The recommended firmware versions are:

- TD520 and TD510 - v4.31.2
- TD540:
  - Partitioned units - v002.000.015
  - Non-partitioned units - v002.001.015

The minimum firmware versions that are compatible with this release are:

- TD520 /TD510 - v4.31.1
- TD540:
  - Partitioned units- v002.000.015
  - Non-partitioned units - v002.001.015

Refer to the *TD5x0 release notes*, available within the TD5x0 OS Release Packages on Partners, for more information.

2.3 Storage space requirement for firmware upgrades

Firmware upgrades require storage space to be available. The amount of storage space required is different for different firmware versions:

- v1.0.x - v2.14.x – at least 1.3 GB of available storage space
- v2.15.x onwards – at least 900 MB of available storage space

When not enough storage space is available, the system deletes old files until enough storage space becomes available. Files are deleted:

- One at a time
- In order by type: <fl2>, <s>, <partial>, <tmp>, <filepart>, <SG6> (at least one SG6 file is maintained), <zsnap>, <Archived_tag>, <Archived_pav>

2.4 Over-the-air (OTA) firmware updates

This section describes remote delivery of firmware updates.

WorksManager can push OTA ECM firmware file (.SG6) updates. This enables a technician to update a machine without being physically present.

When the OTA SG6 file is downloaded, the operator has 2 options:

- Install the OTA SG6 immediately
- Install the OTA SG6 at the next restart

The operator cannot repeatedly postpone the install. During installation, the Operator App and Web Interface are both locked out and show update progress. The SG6 is stored as a temporary file and is removed after installation. Due to the size of the SG6, it may take some time to download. If the download is interrupted, it resumes downloading from the point of interruption.

A technician can select an OTA SG6 file and apply it without waiting for the operator to acknowledge it. When the system restarts after installation, no further auto-install is required because the OTA SG6 is already applied.

When the ECM firmware is updated, individual devices on the system may no longer meet the minimum firmware required. Depending on the device, the system may update the device's firmware so the system can continue operating. See Automatic upgrade of devices.

If the technician installs another SG6 in the presence of the OTA SG6, the OTA SG6 will be applied after reboot because the system detects the downloaded OTA SG6 at start-up. The technician should unassign the OTA SG6 in WorksManager if they want to activate another SG6.

When the update is complete, the system deletes the downloaded SG6 file to free up storage space.

You can still perform system updates by downloading an SG6 file to the system through the Web Interface.

2.4.1 OTA firmware update troubleshooting

If the ECM does not have enough free space, an error message displays. The system requires enough free space for both the SG6 and the files extracted from it. The system needs three to four times the SG6 size to download, uncompress and install the firmware file.

There is a Firmware Upgrades diagnostics section included in the diag.txt with the following information:

- Last downloaded firmware name
- Last downloaded firmware ID
- Time of last OTA server check
- Time of last firmware download
- Time of last firmware upgrade
- Latest OTA upgrade status
- Time of last successful upgrade

If your system is missing an up-to-date maintenance license when a new SG6 is assigned, a warning message displays and the install does not occur.

2.5 Automatic upgrade of devices

When some connected devices do not meet the minimum firmware requirements of the system, the device’s firmware is automatically updated so the system can continue operating. This is generally triggered by:

- Minimum version requirements changing in newer system software
- A newly connected device not meeting that system's minimum firmware requirements

The Operator App and the Web Interface will show firmware update progress and provide the estimated time remaining. During this time, the system's Maintenance Lock is enabled. The technician can choose to override the automatic updates and cancel them. This gives the lock back to the technician.

The operator cannot cancel the updates because they are required to get a Start button with their current configuration and Machine Setup.

The device types that are updated when their firmware does not meet the minimum requirements are:

- SNR radio
- GS51x sensor
- GS520 sensor

- MS9x2 receiver
- MS9x5 receiver
- VM510 valve module

Devices that are not automatically updated and that do not meet the minimum firmware requirement must be updated separately.

2.6 Configure software using the Web Interface

2.6.1 Set up the Wi-Fi network

The EC520-W works wirelessly with any PC, laptop, tablet, or smart phone. Do the following:

1. Set up Wi-Fi:
   a. Select the channel the office uses to communicate with this machine.
   b. Enter your Service Set Identifier (SSID).
   c. Enter a password in Password and Confirm Password fields.
   d. Click Save.
2. If you want to confirm Wi-Fi is connected, open the Settings tab on other tablets or laptops and confirm they are connected to the EC520 network.
3. If required, open the Web Interface on your tablet or laptop and add permission to run the Operator App on your tablet.
   a. Select Configure > Install Assistant > User Permissions.
   b. Select Add to add an operator.
   c. Enter the required details and click Save to save the user.
   d. Tick the checkbox next to the operator to edit or delete the entry.
   e. Toggle off the "Require Operator Password" option to disable operator password while operator log in. This option is toggled on by default.

> **TIP –** For best performance when using Autos-related features, avoid running the Web Interface on a laptop connected to the system via Wi-Fi. Either run the Web Interface on the TD5x0 display or connect the laptop to the system with an Ethernet cable.

Some of the problems that you might see include:

Problem	Description	Resolution
With TD5x0 attached, the Wi-Fi Network page is not complete.	The Wi-Fi network on the EC520 is off, or the EC520 is not enabled for Wi-Fi.	With a laptop connected via Ethernet cable to the EC520 (or via the TD5x0), set up the machine in the Web Interface, then configure the Wi-Fi network as described above.
Wireless tablet cannot communicate with EC520.	The first time you try to communicate with a new EC520 you must be connected either by configuring the Wi-Fi or by using an Ethernet cable. Tablets do not have an Ethernet connection.	Configure your wireless tablet to connect to the EC520 Wi-Fi.

NOTE –The EC520-W can communicate with any Wi-Fi enabled device, but on the machine your TD5x0 display will, by default, use the Ethernet segment of the EC520 long platform harness to connect to the EC520. If the TD5x0 and the EC520 are not connected by an Ethernet cable, then the TD5x0 can be configured to communicate with the EC520 by Wi-Fi. This method of connection is not recommended for Autos-related functions.

2.6.2 Complete the Configure section

In the Web Interface complete all of the Configure section.

1. If using a computer (laptop connected by Ethernet or Wi-Fi), start the laptop.
    a. Open Chrome (web browser).
    b. Enter 192.168.168.1.
    c. Skip to step 3.
2. If using a TD5x0 display, as soon as you supply power it starts and the Welcome screen appears.
3. Tap **Got It** to continue and complete the Android tour screens.
4. Tap the Web Interface icon (shovel and wrench).
5. Log in to the Web Interface using the default username (admin) and create a password.
6. Wait for the initial install wizard to open at the Setup screen and complete the Setup tasks.

> **NOTE –**Using punctuation in the machine name is unsupported and may cause unexpected behavior. Avoid using punctuation in machine names.

7. When finished, the Install Assistant page opens. Acquire the maintenance lock and complete:
   - Install Assistant:
     - Sensor Correction Check - do this check first, if available
     - Measure-up
   - External Lightbars - optional
   - Wi-Fi Network

Things to note:

- Enable the external lightbars. Within a few seconds the lightbars, if connected, flash in sequence. For information on the lightbar start-up sequence, refer to the *External Lightbars* guide in the overflow menu of the Operator App or on Partners.
- Complete the Remote Switches wizard. On dual CI520 interface module configurations, if the CI520s are disconnected from the harness after completing the Remote Switches wizard, redo the Remote Switches wizard to confirm that the remote switches are working as initially configured.

2.6.3 Check GNSS components are connected

After installing the hardware, check the following key components are connected, if required:

- Radio or internet gateway (receiving GNSS corrections)
- GNSS receivers (receiving position from satellites)

Do the following:

1. Open the Web Interface.
2. Select Network > GNSS Correction Source. The GNSS Correction Source screen shows the status of the radios.
3. If the screen shows "Not Connected" status, resolve the problem.
4. Activate and edit the desired GNSS Correction Source, and configure the settings for your site.
5. If the settings will not change, resolve the problem.
6. Select Monitor > Connected Devices, review the GNSS device information, and look for problems. For example, if RTK is not present then the system is not receiving corrections.
7. Resolve any problems.

At this stage your system will have an accurate position and be receiving GNSS corrections.

Some of the problems that you might see include:

Problem	Description	Resolution
The GNSS Correction Source page shows the radio status is "Not Connected".	The grade control system uses the radio or internet gateway device to send GNSS corrections to the EC520. The Correction Source page in the Web Interface shows whether the radio is connected.	• Check the cables between the radio and EC520 are connected. • Check the radio firmware is up to date. Download the most recent firmware from your usual source.
The number of the channel on the radio will not change.	Once you have connected the radio, the one thing left to do is enter the channel that your site uses. If you cannot change the channel this indicates a problem.	Check the radio firmware is up to date.
The GNSS status is not RTK.	Once you have connected the GNSS receiver or receivers, and the site's positioning system is working, the GNSS receiver status will show as RTK. If it does not, this indicates a problem.	Check the positioning system: • Base station • Receiver firmware • Receiver hardware • Cabling
My laptop was connected to an EC520, but I moved and it isn't.	The laptop was connected to an EC520 via Wi-Fi, either directly or through docking station, but moving to another machine or undocking the laptop means the laptop disconnects.	Before taking the laptop to another machine or undocking, try using the keyboard commands Ctrl + Alt + Delete to secure your laptop. Or, reboot the laptop when at the new machine.

Problem	Description	Resolution
In the office my laptop was connected to an EC520, now it isn't.	You may have your laptop set up to connect automatically to the nearest Wi-Fi network. Bringing another EC520 near to your laptop may override the current connection.	In the task menu check the name of the network your laptop connected to. Also found at the top right hand corner of the Web Interface.
Earthworks on supported third party tablet will not appear.	Earthworks is installed but the way you logged in does not have rights to open Earthworks.	In the Web Interface, set -up permissions to allow you to use Earthworks.
'Device Status Error - GNSS Receiver' message displays when starting a cab mount measure-up	On a cab mount configuration, all devices are confirmed as connected but an error message displays when a measure-up is started.	Disconnect any blade mounted receivers and restart the measure-up section. When the measure-up is complete, reconnect the blade mounted receivers.

2.6.4 Customize the MSS band settings - if needed

NOTE –These instructions are for machines using MS9xx receivers with firmware prior to v5.41. If using current firmware, leave the Use Custom toggle off as the latest receiver firmware has the correct frequencies and baud rates built in.

To customize the MSS band settings for receivers using old firmware:

1. Open Operation > GNSS Management
2. Enable the Custom RTX toggle
3. Update the values as required, frequencies listed in Trimble Bulletin

All settings on this page are sent to the MSS receiver or receivers on every system start up.

2.6.5 Start using the TD5x0 display

To set up Earthworks on the TD5x0 display:

1. Configure the system completely in the Web Interface, ensuring an operator is configured in the Web Interface under Configure > Install Assistant > User Permissions.
2. Release the maintenance lock.
3. Supply power to the display.

4. Decide how you want to use the display, landscape (horizontal) or portrait (vertical). Rotate the display to this position. Tap the orientation button on the left side of the screen to adjust the screen orientation.
5. Select the language you want and then tap the arrow in the upper right corner.
6. Select your time zone and then tap the arrow in the upper right corner.
7. Set the date and time by scrolling each item and then tap the arrow in the upper right corner.
8. The Welcome screen appears. To change wallpapers, widgets, and settings, touch and hold the screen background.
9. Tap Got It to clear the Welcome screen.
10. If the display is running an older or newer version of the grade control system to the EC520, then upgrade/downgrade to the matching version.
    a. On the Upgrade Required screen tap Download.
    b. The Downloading percentage screen appears.
    c. The Machine Control Plugin screen appears, tap Install.
    d. On the App Installed confirmation screen, tap Done.
    e. If you are prompted to install an update to the existing application, tap Install.
    f. Tap the Earthworks icon.
    g. When asked to allow the Machine Control Plugin to access files on the display, tap Allow.
    h. Repeat steps a - c.
    i. The **Allow the machine control plugin to access photos, media and files on your device** request appears. Tap Allow.
    j. When you see the Viewing Full Screen panel on display screen, tap Got it.
    k. Enter the name and password set in the Web Interface's Configure > Install Assistant > User Permissions and Login.
    l. The Synchronizing... screen appears.
11. The Dashboard opens.

Earthworks now shows the current machine data. Finish configuring or installing to clear any error states.

2.6.6 Upgrade or downgrade the EC520 firmware

To upgrade or downgrade the EC520 system ECM firmware:

1. Open the Web Interface and browse to Monitor > Onboard Devices.
2. Select + Add Firmware.
3. Browse to the new firmware file, select it, and then select Open. The firmware starts uploading to the system ECM.
4. Wait for the upload to complete, then select the down arrow at the end of the system ECM row.

5. Select the new firmware and then select Apply Firmware. Once the installation completes the system ECM reboots.

> **NOTE –**When the EC520 firmware is upgraded or downgraded, the machine settings files (*.machine) automatically save, preserving previously configured settings even when flashing issues occur. To access the automatically saved files, in the Web Interface, open Advanced > Machine Settings Files.

> **NOTE –**When the EC520 firmware is downgraded, you must apply a machine settings file that is from the same downgraded version or earlier, or recommission the machine.

> **NOTE –**If you want to downgrade from v2.3 or later to v2.0, you must first downgrade to either v2.2 or v2.1 and then downgrade to v2.0.

2.6.7 Upgrade the MS9xx GNSS receiver with the minimum firmware

To use 2.24.x and later of the grade control system, the following GNSS receivers must be upgraded with the minimum firmware:

- MS9x6 - firmware v5.56 or later
- MS9x5 – firmware v5.45 or later
- MS9x2 – firmware v5.31 or later

The latest required firmware is included in the software. After installing the software, in the Web Interface, open the Monitor > Onboard Devices to confirm the minimum firmware is installed.

2.6.8 Upgrade the GS5xx sensor firmware

When upgrading the firmware on a GS5xx sensor, avoid interrupting power to the sensor. Interrupting power may render the system incapable of identifying the sensor.

If power is interrupted, restart the system. If this does not make the sensor identifiable, contact your support channel.

2.6.9 Update the VM510 valve module and CAN Gateway

When upgrading motor graders, dozers or compact loaders from previous VM510 versions, the firmware must be updated to match the requirements below. The firmware is bundled with the software. To update:

1. Open the Web Interface's Monitor > Onboard Devices screen.
2. Select the available firmware for
   a. Valve module - v1.02.10 or later
   b. CAN Gateway - v1.04.00 or later
3. Tap Apply Firmware.

2.7 Tasks

This section describes some common tasks the installer may need to do.

2.7.1 Transferring Files

File Transfer screen

The File Transfer screen is available in the System Settings menu on the operator's interface. You can:

- Synchronize the files on the EC520 controller (the machine) with your organization on Connected Community (TCC).
- Import files to the machine from the display's hard drive or other external storage device.
- Export files from the machine to the display's hard drive or other external storage device.
- Custom file import files to the machine from the display's hard drive or other external storage device.

File transfer from an external storage device to a TD5x0

This section provides useful information about transferring files from an external storage device, such as USB flash drive, to a TD5x0 display.

For detailed information about synchronizing, importing and exporting files, refer to the operator's guide *Transferring Files*.

Requirements

You must have:

- a TD5x0 display
- an external storage device, such as a FAT32 format USB flash drive with a maximum capacity of 32GB

Files are overwritten with newest versions

When transferring files from an external storage device to the machine, all files on the machine that have the same name as files on the external storage device will be overwritten with the external storage device files. This includes the user preferences file userdata.pref.xml, which saves the settings; for example, the text items that are configured.

To keep the user preferences saved on a machine, either:

- Perform a File Transfer of the EarthworksData to the external storage device first, then perform a File Transfer from the external storage device to the machine

OR

- Untick the EarthworksData box when transferring files from the external storage device to the machine

> **NOTE –**The userdata.pref.xml file is stored at ProjectLibrary > EarthworksData > [Machine Name Folder]

Folder structure

When using Import Files TO Machine, the required folder structure on the external storage device is:

- ProjectLibrary/Projects/<SiteName>/OfficeData/Designs

where:

Folder Name	Description
ProjectLibrary	Fixed root-directory folder name.
Projects	Fixed folder name.
SiteName	Name of the job site or project defined by user in the office software.
OfficeData	Fixed folder name. Includes: • coordinate systems (.cal) • site control points (.cpz) • site line work and site points (.smz) • avoidance zones (.avoid.svl) • feature codes (.fxl) • ground surface (.ttm) • current elevation (.cat)
Designs	Fixed folder name. Includes: • design files (.xml, .dsz and .vcl) • elevation design (.cat)

> **NOTE –**Custom File Import TO Machine does not require the above folder structure.

File name rules

The system takes files that are in the OfficeData folder and uses them across the whole project. This includes Depth and Slope and Design guidance modes. The system uses only one file from each OfficeData category— .cal, .cpz, .smz, .avoid.svl, and .ttm. Use these file rules:

- .cal, .cpz and .smz - the system uses the highest version
- .avoid.svl - only one file within OfficeData is supported. Multiple files cause an error message to display on the operator interface when the project is loaded
- .ttm - the operator selects a file within the mapping dialog

File versions

File versions are set by the file transfer method used to transfer files to the OfficeData folder:

- Trimble Business Center - the system automatically adds a version number to the file name, using this format: *'file name'.Vxx.'file extension'*. Example: *NorthArea.V01.cal*
- Other method - the user decides whether to use this naming convention or not.

NOTE –The "V" is upper case.

In the OfficeData folder

In the Projects > OfficeData folder, the system reads both the file extension (.cal, .cpz, .smz) and the number (.Vxx) and then uses the following rules:

Description	Example
The system uses the largest .Vxx number associated with each file extension.	*NorthArea.V01.cal*, *NorthArea.V02.cal*, *South Area.V03.cal* The system uses *South Area.V03.cal*
If the .Vxx number AND the file extension are the same for two different files, the system displays an error.	*SouthArea.V01.cal*, *NorthArea.V01.cal* ".V01.cal" is the same for both files. The system displays an error and uses neither file until the user removes one file.
The system uses a file with a number in .Vxx format before a file without one.	*NorthArea.cal*, *NorthArea5.cal*, *NorthArea.V01.cal* The system uses *NorthArea.V01.cal*
If the file does not include .Vxx in the name, the system does not allow two files to have the same file extension.	*NorthArea.cal*, *SouthArea.cal* ".cal" is the same for both files. The system displays an error and uses neither file until the user removes one file.

In the OfficeData > Designs folder

In the Projects > OfficeData > Designs folder, the system reads both the file extension (.dsz) and the numbering system (.Vxx) and then uses the following rule:

Description	Example
All .dsz files appear on the display and the operator chooses which file to use.	*WestDesign.V01.vcl*, *EastDesign.V01.vcl*, *WestDesign.V02.dsz,EastDesign.V02.dsz*

2.7.2 Export a file or folder to Connected Community

There are two ways to export the Earthworks machine file or folders to Connected Community (TCC).

NOTE –This information applies only to users connected to Connected Community.

Export from Earthworks

1. Capture a backup of the complete system file structure.
2. Open Earthworks and tap Settings > File Transfer > Export from Display.

Export from the Web Interface

Complete one of the following:

- In the Web Interface, select Advanced > Machine Setting Files. Select the machine setting file you want to export, select the expanded menu button (bottom right), and select Download. A machine file has a .machine suffix.
- Capture a backup of the complete file structure along with current settings. In the Web Interface, select Operation > File Management and select Backup All. Upon completion, the backup file will be saved to the Download folder on the device that the Web Interface was being run on when the backup process was started, e.g., if you access the Web Interface via the TD5x0 and initiate the backup from there, on completion the backup file is saved to the Download folder on the TD5x0 and can be retrieved from there.

2.7.3 Convert an existing design file

The existing GCS900 or AccuGrade system uses a different type of design file to Earthworks. Use any of the following methods to set up the machine to use the correct type of design file.

Method 1: Use supported Trimble Business Center objects

Trimble Business Center supports the following Earthworks objects:

- Surface-like objects, including:
  - Surface
  - Corridor Surface
  - Depth Surface
  - Layer-based Surface
  - Over-excavation Surface
  - Planar Surface
  - Subgrade Surface
  - Topsoil Surface
  - Uncompacted PCS Surface
  - Trench Surface
- CAD linework objects, including:
  - Linestring
  - CAD 3D PolyLine
  - CAD PolyLine
  - CAD Arc
  - CAD Circle
  - CAD Line
  - CAD MultiLine Text
  - CAD Text
  - Horizontal Alignment
  - Breakline
  - Polygon
  - Boundary
  - Point and CAD Point - Marked on the background map, but not loaded as a guidable point

Method 2: Use Trimble Business Center

1. Open the existing Business Center Project file (.vce), which contains the existing design(s), using the latest version of Business Center.
2. If the original .vce project is not available:
   a. Best practice is to import the raw design file.
   b. Alternative method is to import the existing GCS900/AccuGrade design. Note that this may cause loss of some design detail such as alignment names.
3. Use Field Data > Publish to Library in Trimble Business Center to export the design for Earthworks vx.x. This will include the .dsz file, along with supporting coordinate system .cal file with GeoData folder, if required.

4. The files will sync with the EC520 at the time interval you set in "Transfer the files to the machine via Connected Community or USB/Display", as per the File Transfer process.

When you have completed the steps you will have a .dsz design file ready to use on the machine.

Method 3: Use File Flipper

File flipper converts GCS900/AccuGrade design files to Earthworks file types so design files can be re-used.

1. Transfer the design files from the CB4x0 onto an external storage device.
2. Use Trimble App Central or Partners to install File Flipper onto the TD5x0 or third party supported tablet.
3. Insert the USB flash drive into the TD5x0 and use File Flipper to convert the legacy design file to the .dsz file type.
4. Store the design in the Projects folder in the office software.

> **NOTE –**Earthworks uses a .cal Site Calibration file that is stored in the OfficeData folder.

Method 4: Custom File Import TO Machine

Custom File Import TO Machine enables:

- File selection from an external storage device with any file folder structure
- Automatic conversion of .svd and .svl files to .dsz

When importing .svd and .svl files, both files must:

- have the same name
- be stored in the same folder

The Custom File Import TO Machine process then converts the .svd and .svl and creates one .dsz, which is then stored on the machine.

2.7.4 Configure an avoidance zone

Follow these rules to configure avoidance zones:

- Avoidance zones must be closed polygons.
- The maximum number of edges of all polygons in a file is 800.
- Export .avoid.svl files from Trimble Business Center using Machine Avoidance Zone exporter.
- Save .avoid.svl files to the OfficeData folder.
- Some measurements for machine boundaries in relation to avoidance zones are estimated. Confirm or update these measurements in Configure > Install Assistant > Additional Machine Dimensions.

> **NOTE –** When an avoidance zone file has an error or warning, the operator can use the associated project and design files but avoidance zone guidance is not provided.

2.7.5 Start working on a supported third party tablet

Read the guide describing installing a supported third party tablet onto a generic excavator system, *Supported third party tablet install guide*.

2.7.6 Reset software to default settings

In certain circumstances you may need to complete a full factory reset of the EC520. Best practice is to Reset to Default when changing machine types or when downgrading the EC520 software. When permanently removing a Earthworks system, Reset to Default effectively clears all stored data.

Before starting, make a copy of all of the files in the file tab, or export a full copy of the machine files from the Earthworks app.

1. Open the Web Interface and select Advanced > Reset to Default.
2. Read the Reset to Default warning message, ensure your files are backed up and select Reset.
3. Type 'reset' into the Confirm Reset Of All Data, User Accounts, And Machine Measurements box and select Confirm Reset.

> **NOTE –** Resetting to factory defaults does not affect licenses.

4. After resetting to factory defaults, the Web Interface opens:
    a. Enter a new password
    b. Accept the End User License Agreement
    c. Configure the machine again

2.7.7 TD5x0 Recovery Mode

Recovery Mode

Recovery Mode is an option on the power down menu. When opened, Recovery Mode provides an option to clear a TD5x0 hard drive. This option is useful for resetting a TD5x0 password.

Reset a TD5x0 password

To reset the password on a TD5x0

1. Turn on the display. Press and hold the power button to open an options menu
2. Tap Recovery and then OK on the Recovery Mode box
3. Wait for the No Command, or dead Android, image to display
4. Press simultaneously the power and volume up button a few times until the Android Recovery menu opens

5. Follow the on-screen instructions to select wipe data/factory reset
6. Wait for the display to reboot and enter a new password

> **NOTE –**Wipe data/factory reset erases all files, such as design files, from the display's hard drive. All installed apps remain.

Reset a TD540 Android password

To reset the Android password on a TD540:

1. Turn off the display. Press and hold the power and volume down buttons at the same time until the display turns on.
2. Wait while the device image displays.
3. Press the volume up button for five seconds and release. This makes the menu open.
4. Follow the on-screen instructions to select wipe data/factory reset.
   Note - Tapping the screen also scrolls through the menu items.
5. Wait for the display to reboot and enter a new password.

> **NOTE –**Wipe data/factory reset erases all files, such as design files, from the display's hard drive. All installed apps remain.

Exit the No Command screen

To exit the No Command, or dead Android, screen (before the Recovery Mode menu screen), press the display's power button or leave the display on for two minutes and it will automatically reboot.

2.7.8 Exit Recovery Mode menu

To exit the Recovery Mode menu, follow the on-screen instructions to select reboot system now.

2.7.9 Configuring a serial radio

The standard system radios are SNRxxx models. However, the system also supports generic serial radios for communication. If using a generic serial radio, configure it as follows:

Prepare the Generic (Serial) radio (required):

- Use the radio's own software to configure the radio separately. The Earthworks system cannot change a generic radio's settings.
- Set the baud rate to 38400 bits/second.
- Set the serial connection to 8 data bits, no parity, and 1 stop bit.

Enable the Generic (Serial) radio option in the system using the Web Interface (optional):

- If the system detects data on the auxiliary serial port (for example, corrections), it automatically enables the Generic (Serial) radio.

If desired, you can manually enable the Generic (Serial) radio option. From the Web Interface's Configure > Install Assistant screen, proceed through the Setup section and enable the Generic (Serial) toggle.

Select the Generic (Serial) radio as the active correction source in Earthworks using either the Operator App or the Web Interface (required):

- In the Operator App, select the Generic (Serial) radio as the active Correction Source from the Machine Setup tile.
- In the Web Interface, activate the Generic (Serial) radio on the Network > GNSS Correction Source screen.

2.7.10 Configuring a generic Ethernet gateway

The system supports generic Ethernet gateways. If using a generic Ethernet gateway, configure it as follows:

- Connect the generic gateway device to the system harness using the Internet ETH2 connector.
- Use the gateway device's own software to configure the device separately. The Earthworks system cannot change a generic gateway's settings.

Enable the generic gateway device in the system using the Web Interface:

1. From the Web Interface's Configure > Install Assistant screen, proceed through the Setup section to the Network devices screen.
2. Set the *Choose a device configuration* drop-down to Generic Gateway (Ethernet).

The following conditions apply:

- If the system is configured for an internet-based GNSS correction source, the generic Ethernet gateway is a required device.
- If the system does not require the internet for guidance, the device is listed as optional.
- If the system does not detect any Ethernet traffic, it lists the gateway device as missing.

2.8 Set up licenses for this machine

2.8.1 Introduction to licenses

A Software Maintenance License must be installed to use the system. You can manage licenses on the Web Interface's Advanced > Licenses screen. There are two types of Software Maintenance Licenses:

- Perpetual – an open-ended license. Stored on the EC520 controller or the TD5x0 display, and not transferable to another device.
- Subscription – a license that expires after a fixed term, if not renewed. Stored on the TD5x0 display or the GNSS receiver and not transferable to another device.

You can install licenses manually on your system, or sync them with the cloud.

The system provides a status icon beside each license:

- Red cross: You cannot operate the system with the current licenses. A license that is required to access the work screen is missing or expired.
- Orange exclamation mark: You can operate the system, but functionality may be limited or due to expire. Options include:
  - The license is currently usable, but is within its expiration warning period.
  - For the current configuration, a module license that is expected, and which adds functionality, is missing or expired. You can still access the work screen.
- Green tick: The license is installed correctly.

2.8.2 Apply required core licenses

The following core licenses must be applied before starting the following valve calibrations:

- Valve calibration: Core License - Motor Grader

2.8.3 Apply a Software Enabled Attachment license

Apply a Software Enabled Attachment license on machines with a 3D license.

2.8.4 Updating licenses with an internet-connected machine

> **NOTE –**Your TD5x0 display must have firmware V4.30.0 or later.

If you have an internet connection, there are two ways to sync the licenses for your system with the cloud. Both options are on the Web Interface's Advanced > Licenses screen:

- If you enable the Auto Sync toggle, the system automatically synchronizes the licenses with the cloud every minute.
- If you select the Sync Now button, the system connects to the cloud and checks for new licenses once.

If the licenses installed on a device are from the cloud, a cloud icon shows beside the device . Once a device has cloud licenses loaded and this icon is present, it can no longer load or use previously-applied option keys. The Licenses screen shows the license synchronization status at the top of the screen.

Check that your licenses appear correctly.

> **NOTE –**When you apply a subscription license to an MS9xx receiver, you cannot downgrade the firmware below version 5.54.

2.8.5 Updating licenses manually

When you manually install a license file:

- The system installs the license on the device covered by the serial number in the license file name.
- The file name is case sensitive. Ensure the extension is lowercase.

To install the license:

1. Make the license file accessible to the computer or device you are using to enter data: via a USB flash drive, on the device or on your network.
2. Open the Web Interface and select Advanced > Licenses.
3. Select Add License File and browse to the license file.
4. Select Add.
5. In the Advanced > Licenses screen, check that your licenses appear correctly.

2.8.6 License errors

For information on license error codes, see License troubleshooting.

2.9 Online services

Depending on your system's configuration, you can connect to an online service:

- Trimble Construction Cloud
- VisionLink

An online service may provide benefits like automatic updates of projects/designs or automated system firmware updates.

2.9.1 Configuring an online service

To configure the connection to an online service, use the Web Interface's Network > Cloud Services screen.

Trimble Construction Cloud

Trimble Construction Cloud supports the following features:

Field	Description
Service Connection	Reports the status of the latest sync with the online service: • Green ticket: The most recent sync was successful • Grey icon: The system has not synced with the online service yet • Orange warning: The most recent sync failed Select the icon to manually resync with the service.

Field	Description
WorksManager Reporting	Enables reporting position and status data to WorksManager.
WorksManager Firmware Over The Air	Enables WorksManager sending assigned firmware updates to the grade control system.
WorksOS Surface Download	Enables the operator to download a surface file (.ttm) from WorksOS: • In the Operator App's Cut/Fill Mapping screen, the operator can set the Ground Surface drop-down to WorksOS Download. • The machine's EC520 must be a registered device in your WorksOS account. • Your WorksOS account must have an active project for the machine's location.
Report Files (.tag)	Enables sending .tag files to WorksOS and VisionLink. If you disable this toggle, the .tag files are saved locally.
Remote Access	Enables remote access to the Web Interface.

VisionLink

VisionLink supports the following features:

Field	Description
Service Connection	Reports the status of the latest sync with the online service: • Green ticket: The most recent sync was successful • Grey icon: The system has not synced with the online service yet • Orange warning: The most recent sync failed
Remote Access	Enables remote access to the Web Interface.
Report Files (.tag)	Enables sending .tag files to VisionLink. If you disable this toggle, the .tag files are saved locally.

2.9.2 Additional options

The Additional fields enable the system to send anonymous reports about your machine to maintain, support and improve the software.

Field	Description
Crash Reports	Crash Reports contain crash and bug data about your software.
Usage Statistics	Usage Statistics send anonymous data about how operators use the app. This data helps to improve future releases.

2.9.3 WorksOS errors

See WorksOS errors.

2.9.4 Communication with Cat online services

If the grade control system is connected to an A6N2 network manager (with the latest firmware: P/N 6467493-00.fl2) and a PL radio, the system will check that a valid Cat subscription exists. If the subscription is valid, the system can communicate information with VisionLink. This can extend some functionality.

If the system detects a base level subscription, the 4 core pieces of data that the system reports to the online service are:

- System is operating
- Power Down Delay
- License info
- PL radio signal strength

If the system detects a Cat Performance + Grade Connectivity subscription or a Performance Pro subscription, it also enables:

- Remote Access
- Cell GNSS corrections (VRS), if that is the configured correction source

If the system detects a Cat Performance Pro subscription, it also enables:

- Project Data:
  - Designs
  - Site Calibrations
  - Geo Data
- Tag files
- System snapshot
- Status Message (new)

If either of the additional Cat subscriptions is present (Performance + Grade Connectivity or Performance Pro), you will be able to use Remote Access through both VisionLink and Trimble Construction Cloud, when the Primary Cloud Service is set to Trimble Construction Cloud.

The following table shows which online service features are available, depending on the gateway combinations and the Primary Cloud Service that is set:

Gateway Type	Additional Cloud Subscriptions	File Management	Tag Files	Remote Access	Status Message
Primary Cloud Service = None					
None	Performance + Grade Connectivity	N/A	N/A	Cat	N/A
None	Performance Pro	N/A	N/A	Cat	N/A
Primary Cloud Service = Trimble Construction Cloud					
SNM94x radio	-	Trimble	Trimble	Trimble	Trimble
SNM94x radio + PLE702 A6N2 network manager	-	Trimble	Trimble	Trimble	Trimble
PL radio + PLE702 A6N2 network manager	-	Trimble	Trimble	Trimble	Trimble
PL radio + PLE702 A6N2 network manager	Performance + Grade Connectivity	Trimble	Trimble	Trimble + Cat	Trimble
PL radio + PLE702 A6N2 network manager	Performance Pro	Trimble	Trimble	Trimble + Cat	Trimble
Primary Cloud Service = VisionLink					
PL radio + PLE702 A6N2 network manager	Performance + Grade Connectivity	Cat	Cat	Cat	Cat
PL radio + PLE702 A6N2 network manager	Performance Pro	Cat	Cat	Cat	Cat

2.10 Receiving projects and designs from WorksManager

If you are connected to WorksManager, the service can push the latest project and design files to your system. Using WorksManager requires a paid subscription.

When enabled, the grade control system checks WorksManager for updates at start-up and automatically downloads any updates. If there are any subsequent changes to the files, the service sends them to the grade control system. The operator can also manually check for updates by tapping the Refresh button on the Job Setup screen.

> **NOTE –** The version of the project and design on WorksManager is the controlling version, so you must use WorksManager to publish, update, or delete the project or design. If you attempt to modify files on the grade control system, they will be restored to their online state by WorksManager.

Your system can have a combination of local projects and projects from WorksManager. If a project or design is from WorksManager, a cloud icon appears beside it on the Operator App's Job Setup screen. The cloud icon states are:

	For a project: All the project files, including design files, are successfully downloaded. For a design: The design is successfully downloaded from WorksManager and is available to use. The design name will include the version number.
	For a project: Part of the project is still to be downloaded. For a design: The design on WorksManager is new or updated, and is queued to download to your system. It is not currently available to select. If a number appears beside the icon, it represents the design download percentage.
	There was an error downloading the project or design. It is not available to select.
No icon	The design is not from WorksManager.

The download process performs a check to ensure designs are valid.

2.10.1 Troubleshooting

Viewing warnings and errors

To view errors or warnings encountered while connecting to an online service, see the Web Interface's Monitor > Program Log screen.

Viewing the state of published files

You can view the state of published files in WorksManager.

Design validation

If a downloaded file is invalid, the Operator App reports an Issue > ....message.

There is an issue with the file in WorksManager. Check that the file is valid and that the design elements are supported in the grade control system.

For more information on the validation process, see Design validation.

Perform a TCC sync

For new EC520 devices, perform a TCC (Connected Community) sync to authenticate the device before you can use WorksManager.

WorksManager updates project/design that operator is using

If WorksManager updates a project that the operator is currently using, the Operator App reports Project updated remotely.

If WorksManager updates a design that the operator is currently using, and auto-archive is enabled, the grade control system deletes the design and downloads the new version. The Operator App reports Design deleted remotely.

The dashboard displays so the operator can select the updated project/design. If Autos is engaged, the system waits until Autos is disengaged before returning to the dashboard.

Project or design renamed on WorksManager appears to be deleted

If a project or design is renamed in WorksManager, the grade control system believes that it is deleted. The system deletes the local copy and downloads the new version. The Operator App will report either Project deleted remotely or Design deleted remotely.

Download connection is poor

If the download connection is poor, causing low download speed, the operator can select the project and design in the Operator App's Job Setup screen. This prioritizes these downloads above other projects and designs in the queue.

Download fails to complete

If a download fails to complete in the allotted time, the partial download can resume when the internet connection is back online.

2.10.2 Known issues

TCC sync delays WorksManager design download

An issue has been reported where the user could not apply a new design, despite download status of 100%.

The grade control system delays installing a design if it is simultaneously syncing large files from TCC. The design becomes available after the TCC sync completes.

2.11 Syncing files with VisionLink Productivity

If you have a configured account, you can sync files between the grade control system and VisionLink Productivity.

> **NOTE –**VisionLink Productivity requires a paid subscription.

The supported file types are:

- Files sent from VisionLink Productivity to the grade control system:
  - Projects and designs (.dsz, .vcl, and .cal files)
- Files sent from the grade control system to VisionLink Productivity:
  - .tag files
  - System snapshots

This process requires that your grade control system is connected to an A6N2 gateway device.

Infield designs are not synced to VisionLink Productivity.

2.12 Using coordinate systems

GNSS systems initially provide spherical coordinates (positions in terms of latitude and longitude). For example, GPS uses a geodetic datum called *World Geodetic System 1984*.

A coordinate system converts these spherical coordinates into the plane-rectangular coordinates (positions in terms of Northings and Eastings) used in ordinary plane surveying. This coordinate transformation is called a projection. Different projection methods introduce different types and magnitudes of distortions.

To correct these distortions in initial projections, you can apply a coordinate adjustment. The adjustments are typically used to modify global coordinates to achieve a best fit with regional coordinates. The coordinate adjustments may be published by a government agency. For example, adjustments can be contained in a grid file, such as a Geoid Grid File (.ggf) or a Datum Grid File (.dgf).

So, the process is:

1. Spherical coordinates available from GNSS
2. Coordinate system converts spherical coordinates to plane rectangular coordinates
3. Local adjustments are applied via grid file

Key coordinate files

The file CoordSystemDatabase.xml is a database that includes published grid files.

The system uses CoordSystemDatabase.xml to generate a proprietary file called a .cal file. The .cal file is a site calibration file that tells the system the coordinate system used for a project, and any grid files for local coordinates.

In previous versions of the software, if you wanted to use a custom grid file, you had to add it to CoordSystemDatabase.xml, regenerate the .cal file, and place the grid file in the *\GeoData* folder.

Selecting a coordinate system

When an operator creates a project, they can select the coordinate system to work with - if you grant permission (this feature is recommended for advanced operators and is disabled by default). You can enable it via the Web Interface's Operation > File Management screen.

They can either:

- Create the .cal file for the project by selecting the correct coordinate system (such as a national reference system), or
- Copy the .cal file from another existing project.

This feature can be useful if they don't have a .cal file, but know the published coordinate system requirements and have the right correction source.

For more information on how an operator can use this feature, refer to the *Project and Design Files Guide*.

Using a grid file that isn’t published

In previous versions of the software, CoordSystemDatabase.xml and the .cal file must both list any grid files. If they were not in the .cal file, they wouldn’t be usable. These grid files that are referenced in CoordSystemDatabase.xml are called *published*. This made it challenging to add new grid files to the system.

However, you can now also add a grid file to a project without needing to mention it in the .cal file. These grid files are called *independent*. Place the file in the *\GeoData* folder.

2.13 Configure the HMB Controller

Configure the external HMB controller to ON to enable Autos. Once enabled, the ON configuration persists and does not need to be enabled again.

To configure the external HMB controller:

1. Press the right most button under the wrench icon on the machine display to open the service menu.
2. Select ON under External Controller.
3. Exit the service menu.

Autos can now be enabled in the Operator App.

3

Grader sensor calibration and measure-up best practices

In this chapter:

3.1 Overview of sensor calibration

A poorly calibrated sensor introduces large, usually unmeasured, errors into the system. These calibration instructions are intended to enable high accuracy grading over a wide range of conditions.

To calibrate the fixed sensors, use a tool external to the machine and more accurate than the measurements you are trying to obtain. These instructions assume the use of a total station.

Recalibrate the sensors when a sensor is moved or replaced.

3.1.1 Prerequisites

Before starting a sensor calibration:

- Validate the installation of the system components and cables.
- Review this entire guide. This will ensure that you understand the full calibration workflow.

3.1.2 Equipment list

- A device, for example a Trimble TSCx Data Collector, running Trimble SCS900 or Siteworks software, with a fully charged battery.
- One of the following total stations, in good adjustment, and with a fully charged battery:

  - Trimble SPS Series
  - Nikon M series
  - Trimble TS Series
- A builder's level
- Cleaner/degreaser
- Cleaning rags
- A marker pen – choose a color easily visible against the color of the machine
- Self-adhesive acrylic targets
- A steel straight edge
- A carpenter's square
- A notebook and pen
- These instructions

3.1.3 EC520 placement

> **NOTE –**The EC520 ECM must be installed in accordance with the mounting instructions for the specific model of grader blade attachment being used.

The location of the EC520 contributes to the measure-up. If the EC520 is moved or replaced after starting this process, repeat the body sensor calibration.

3.1.4 GS51x on the yoke

Use GS51x IMUs only on the yoke sensor mounting. The yoke sensor mounting was designed for GS51x IMUs only and provides the appropriate bracket structure and yoke clearance.

> **NOTE –**Using an IMU that is not a GS51x may result in unknown performance and damage issues.

3.1.5 Electric mast support

The system supports the EM410 and EM415 electric masts. It does not support the EM400 electric mast.

Minimize mast lean

Always keep the electric mast as vertical as possible. When blade pitch/roll is required, keep the roll within ± 10° from vertical.

For machines measured-up with the multiple bolt hole option enabled, when the blade is pitched/rolled forward or backward to a mast lean exceeding ± 10°, change the bolt hole on the machine and in the Operator App to return the mast lean to within ± 10°.

Whenever you attach a mast to a blade, always push the mast hard forward on the bolts and then tighten the bolts. This puts the mast is in a position that is repeatable for machines with the multiple bolt hole option enabled and for whenever a mast is temporarily removed and then reattached.

> **NOTE –** Mast leans exceeding ±10° from vertical can introduce errors from the sag in the M4 shock mount rubber isolators. Old and worn rubber isolators can also increase sag.

> **NOTE –** Only one measure-up with the correct bolt hole is required. The system adapts guidance for the other bolt holes.

> **NOTE –** Insert a blanking screw into one of the two bolt holes of the lower section of the mast mount before measuring up. If the blanking screw is moved to the other bolt hole after measure-up, a new measure-up is required.

3.2 Check machine wear

3.2.1 Introduction

This section describes the points of the machine that must be checked and adjusted to provide optimal grade control results. If the machine is loose, the grade control system cannot produce optimal results.

For optimal results, you must maintain on an ongoing basis:

- The machine's optimal working condition
- The accuracy of the machine components

3.2.2 Position the machine

Park on a hard, planar surface as near to level as possible. A workshop concrete floor is an ideal calibration pad.

If needed, put blocks under the cutting edge so the target centerline can line up with the edge of the cutting edge without interfering with the surface underneath.

There must be enough space around the machine to allow the machine to be accurately turned 180°.

3.2.3 Check the cutting edge

The cutting edge is defined by points near either end (the recommendation is to measure the cutting edge length 0.3 m/12 inches in from each end), so it is acceptable if the blade has greater wear at one end than the other. However, the cutting edge must form a straight line with a maximum horizontal deviation from a string line of 0 ± 0.005 m (0.2 inches).

If the wear on the blade is inconsistent or the cutting edge is bent (in the forward/back direction), system guidance and cut surfaces will be inaccurate.

3.2.4 Check the circle play

Adjust the circle shims and/or circle wear strips to minimize play in the circle. Follow the manufacturer's instructions and recommended tolerances.

Follow the machine manufacturer’s instructions for adjusting the circle wear strips to minimize horizontal slop in the circle and to make sure the circle is centered.

3.2.5 Check the circle backlash

This is the maximum permitted movement in rotating the circle before the rotation sensor sees a change in direction. The backlash must be less than 12 mm (0.5 inches) when measured on the outside of the circle.

Remove all play from the connection between the hydraswivel and the hose tray.

Remove any horizontal play from the hose tray slide rod guides on the back of the circle.

To minimize the circle backlash:

1. Rotate the circle clockwise about 10° and place a mark on the side of the circle relative to a fixed point on the drawbar.
2. While viewing the RS400 diagnostics screen, slowly rotate the blade counter-clockwise. Stop rotating as soon as the sensor reading begins to change. Place a second mark on the side of the circle at the new position.
3. Ensure the distance between the two marks is less than 12 mm (0.5 inches) .
4. If the distance between the marks exceeds the recommended distance:
    a. Ensure that the circle shoes and shims are adjusted as recommended. See the machine’s operation and maintenance manuals for details.
    b. Tighten the hydraswivel and hose tray assembly. Pay attention to the bolt or pin under the hydraswivel that connects it to the hose tray and check that the hose tray cannot move sideways along the bolt. Pack any spare space with washers or tighten the bolt to remove play.
    c. Check where the hose tray guides are located on the outside of the circle. There should be no sideways play.
    d. If there is sideways play, weld some flat bar on either side of the rods. Be careful to ensure that the rods can still move freely up and down as well as in and out, but not sideways.
5. Once these remedial actions have been done, repeat steps 1 to 3.
6. If still outside tolerance, then check through all of the mechanical connections and remove any play that you can.

3.2.6 Check the lift cylinder backlash

Ensure that the distance between the blade and the circle has a maximum of 3 mm (0.1 inches) of play when the blade is lifted off the ground.

3.2.7 Check the lift cylinder ball joints

Measure the distance between the cylinder rod and the top of the circle in the following situations:

- The circle is resting on the blade; and
- The blade is hanging from the circle

Ensure that the measurements differ by less than 1 mm (0.04 inches).

3.2.8 Check the blade rail shims

Ensure the maximum clearance between the moldboard and wear strips is 1 mm (0.04 inches).

Tire wear and pressure have a large impact on accuracy. Uneven pressure and wear can affect the mainfall calibration.

Ensure that the tires on the tandems are all of the same make, model, and size, and the wear is approximately even:

1. Check that the tread is aligned in the correct direction and that the tires are all inflated equally. The operator must maintain this air pressure to grade with high accuracy.
2. Repeat this process for the front tires.
3. Park on a flat, level surface and measure the height of the wheel rim from the surface. Confirm that the height of the rim is the same between pairs of tires.

3.3 Prepare for sensor calibration

Careful preparation of the machine is needed in order to perform accurate calibrations of the critical sensors. To prepare the machine for calibration, complete the tasks described in the following sections.

3.3.1 Zero the articulation (cross slope (2D) and single 3D systems)

For systems using cross slope (2D) or single 3D operation, it is very important that the machine is straight while making finish grade passes. Make every effort to maintain zero articulation, because articulation causes drift during turns and when working on slopes. If the machine is equipped with a return-to-center feature, use the feature regularly during finish grade runs. If the operator agrees, use an articulation lock pin to avoid articulation.

For systems using dual blade-mount GNSS, articulation is acceptable.

> **NOTE –** To calibrate the “Return Articulation to Zero” feature, if one is installed, follow the manufacturer’s calibration procedure.

To check zero articulation, there are two string line methods:

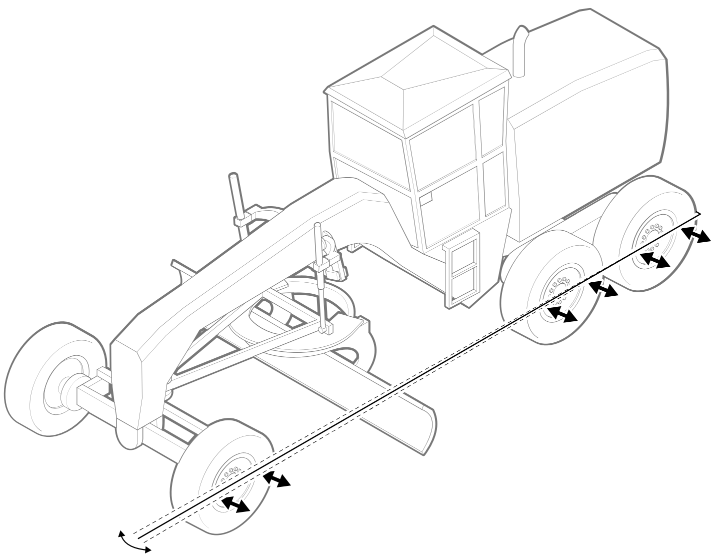
- Outside the wheels
- Inside the rear tandem wheel frames

> **NOTE –**Only make small changes to the machine's articulation when the machine is stationary. If you do change the articulation of a stationary machine, drive the machine backward and forward about one machine length prior to measuring the articulation to release any tension in the tires and linkage.

Using a string line outside the wheels

This procedure assumes that the width between the front wheels is equal to the width between the rear wheels. In addition, the front wheels must be parallel to the goose neck, that is, pointing directly ahead.

1. Park the machine on a flat, level surface.
2. Fix the string line to the back of the right rear wheel at the same height as the axle.
3. At the front wheel, hold the string line at axle height.
4. Measure the offsets from the string line to the wheel rims on the rear wheels. Move the front end of the string line in or out as required until the four wheel rim offsets at the rear are equal. When all offsets are equal, the string line is parallel to the rear wheels.
5. Measure the offsets to the front wheel rims.

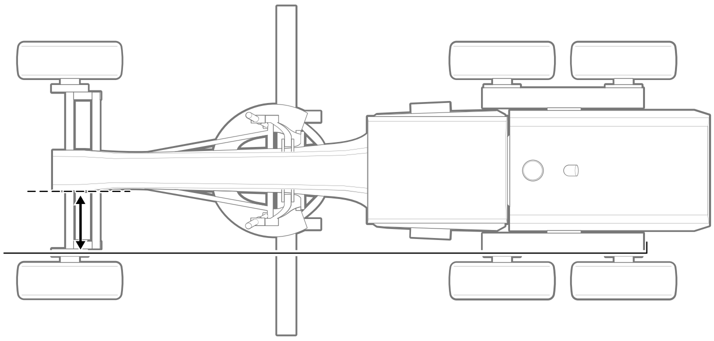


6. Repeat the procedure on the left side of the machine. When the front offsets for each side of the machine are 0 ± 20 mm (0.8 inches), then the articulation is considered to be zero.

Using a string line inside the rear tandem wheel frame

This method is more difficult but more accurate. It may be difficult to find a straight path for the string line past the circle and blade. If you can adjust the circle and blade to be out of the path of the string line on both sides of the machine, use this method.

1. Park the machine on a flat, level surface.
2. Fix the string to a point behind the machine.
3. Pull the string line over the rear tandems so the string is parallel to the tandem frame on the inside of the wheels.
4. Pull the string line past the inside of the front wheels and secure it to a point in front of the machine.

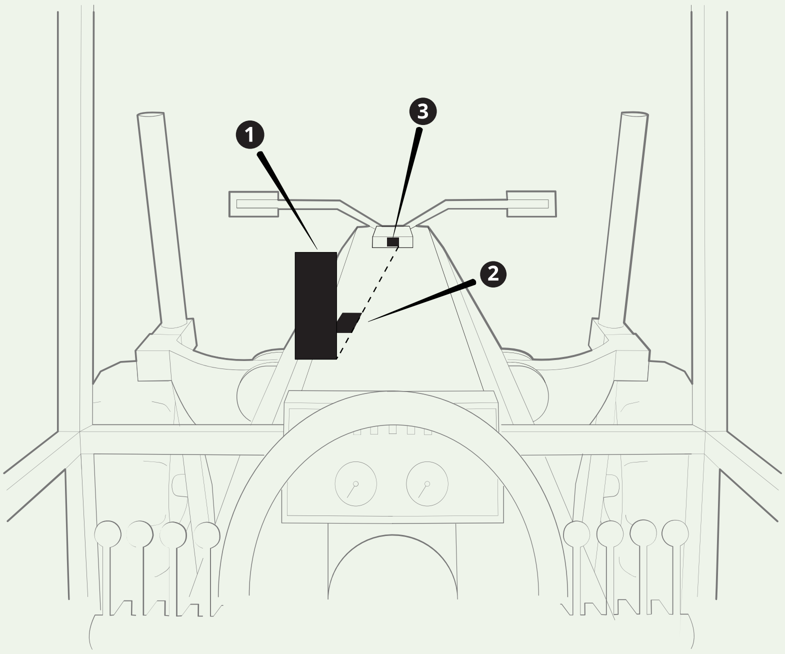


5. Measure the offset from the string line to a part of the front frame that is symmetrical about the center line.

> **NOTE –**The front counterbalance is bolted on and may not be directly in the center.

6. Repeat the procedure on the other side of the machine. When the front offsets for each side of the machine are within ± 20 mm (0.8 inches), then the articulation is considered to be zero.

TIP – On graders where the goose neck pivots in front of the cab, once zero articulation has been achieved, find suitable places on the windscreen (❶) and goose neck (❷ and ❸) for the three marks shown below and place marks in those positions.



❶ – a piece of tape placed on the inside of the windscreen.
❷ – a black paint mark on the end of the goose neck nearest the operator.
❸ – a black paint mark on the goose neck farthest from the operator.

As the marks are very wide compared to the required accuracy, make sure that the right hand side of the marks lines up, and not the centers. During operation, when the operator aligns the right hand side of the three marks, they know that there is zero articulation.

3.3.2 Center the linkbar

For the initial calibration, center the linkbar. To do this, move the locking pin to the center hole.

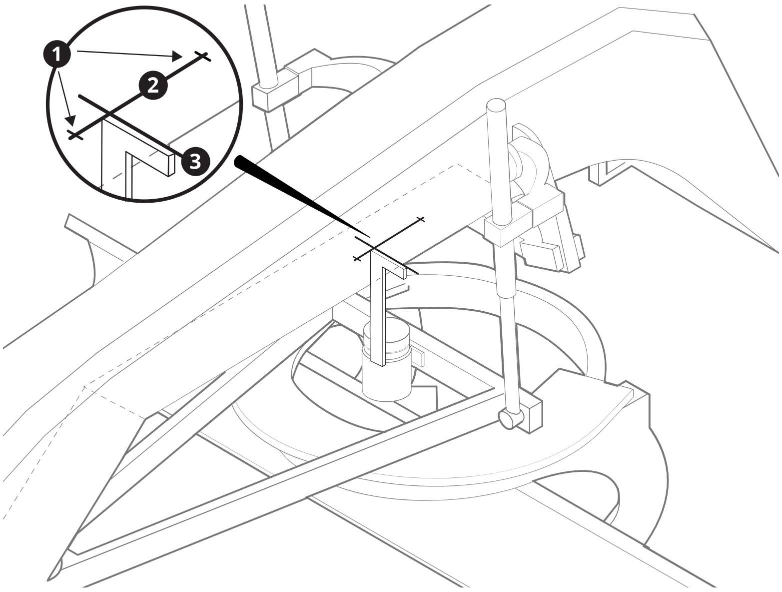
3.3.3 Center the drawbar and create a reference

Center the drawbar

1. Place the blade on the ground in the working position.
2. Mark two points (❶) about 30 cm (12 inches) apart on the bottom of the goose neck. The points must be:
   a. On the center line of the goose neck – use a tape measure to locate the center line

   b. Above the rotation sensor

3. Use a straight edge to draw a line down the center of the goose neck between the 2 points (❷).

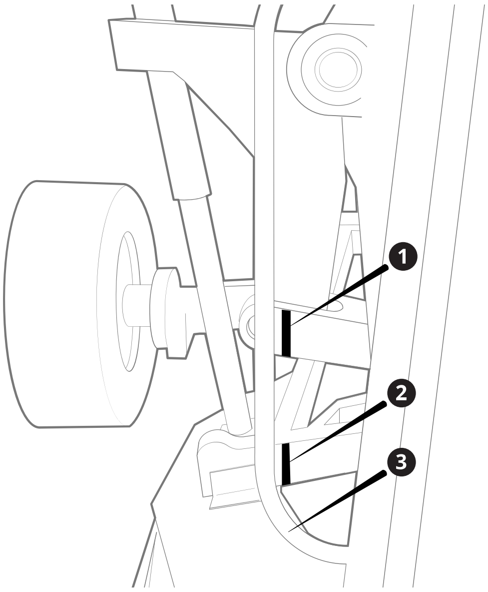


4. Use a carpenter's square to draw a line perpendicular to the center line and directly above the front of the center line of the rotation sensor (❸).
5. Place the carpenter's square flush to the bottom of the goose neck, and align the edge to the cross-hair created.
6. Keeping the blade as close to the working position as possible, side shift the drawbar until the centerline of the rotation sensor aligns with the edge of the carpenter's square.

Create a reference for the operator

To allow the operator to easily center the drawbar:

1. With the drawbar centered and in the working position, the operator in the cab should choose an easily visible, fixed, vertical feature on the machine within the line of sight to both the drawbar and the upper frame. Examples of existing features could be a handrail or machine cover, or a feature could be created by attaching a vertical taped line on the cab window.
2. The operator directs a second person to mark the upper-frame (❶) and the drawbar (❷) so that the marks align with the selected vertical feature, for example a handrail (❸).



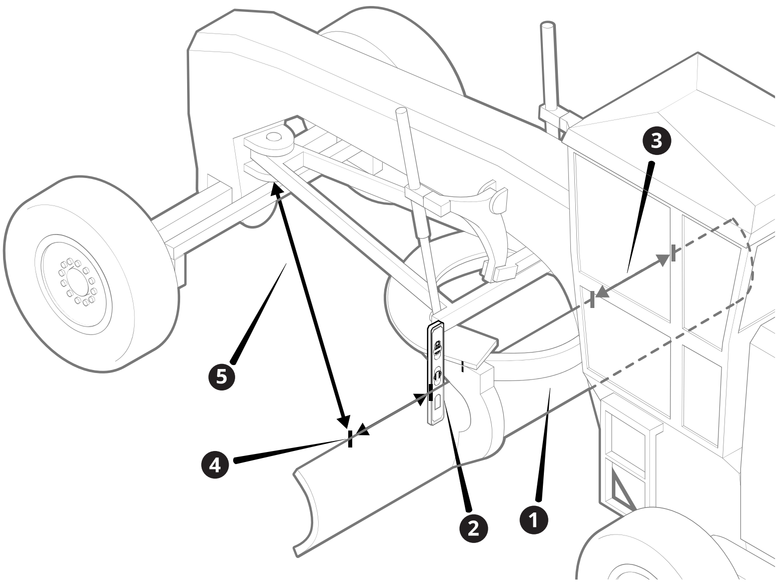
3. Now the operator has drawbar centering guides visible from the cab that can be used during operation to keep the circle centered under the goose neck.

3.3.4 Zero blade rotation

NOTE – This level of accuracy is only required for the sensor calibration. The measure-up only requires the blade to be square by eye to the machine.

You will need a marker, a straight edge, and a steel tape measure.

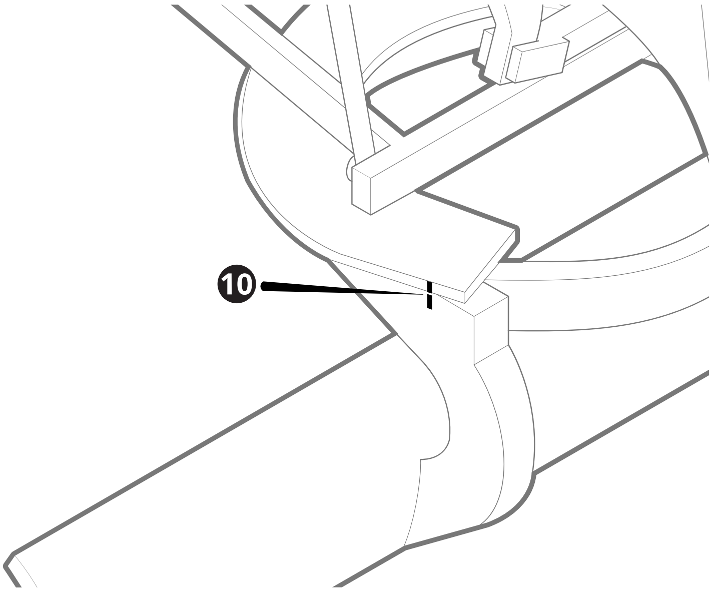
1. Side shift the blade so that it is approximately centered (❶).
2. Using a straight edge on one side of the blade mounting frame, project the edge of the mounting from the circle down to the top edge of the blade. Make a mark on the blade with a marker (❷).



3. Repeat step 2 for the other side of the mounting frame.

> **NOTE –**It does not matter whether the marks are different distances from the *outer* edges of the blade.

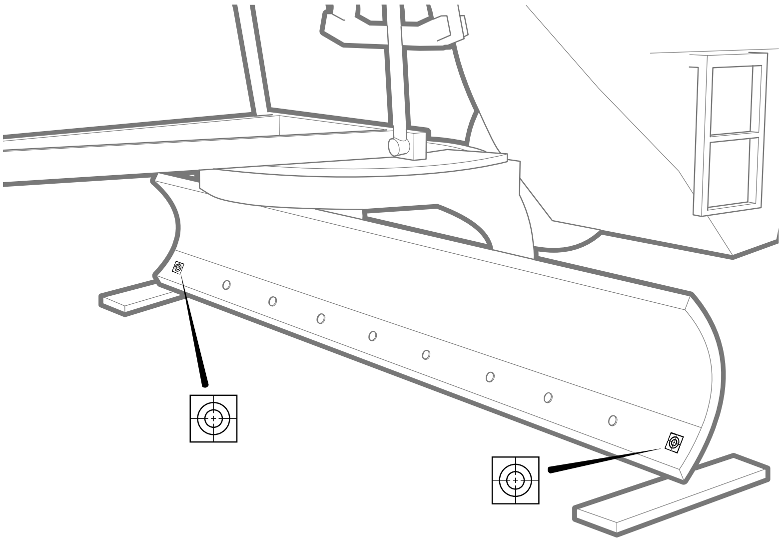
4. From one mark on the blade, measure a fixed offset outward, toward the blade tip on the side closest to the mark, and make another mark (❹). Repeat this for the other side of the blade (❸). The offset on each side must be equal, and the offset distance should be as long as possible. This allows room to use a steel tape to make the measurements in the next steps.
5. From each of these outer marks, measure to a point on the drawbar and on the same side as the mark on the blade (❺). The points on the drawbar must be symmetrical about the centerline of the drawbar, and to the rear of the ball joint.
6. If required, correct the blade rotation and repeat the measurements until the measurements are equal on both sides.
7. When the blade rotation is zeroed, place a mark or reflective tape on either side of the circle in locations that align with a fixed component, and that are easily visible to the operator (❿). This allows the operator to accurately square the blade in the future.



3.3.5 Level the blade (cross slope (2D) systems only)

Calibrating the blade slope sensor with a calibrated digital level can result in errors of up to 30 mm (1.2 inches) across the width of the blade. For best accuracy, use a total station and a device running Trimble SCS900 or Siteworks software, or similar, to record the blade tip positions.

1. With the blade perpendicular to the machine body, use a total station to record the position of the outermost cutting edge bolts at each tip.



2. Adjust one blade tip up or down as required to match the elevation of the other tip within 2 mm (0.1 inches).

Avoid using the cutting edge as a guide to level the blade. The edge may be a different height at each end.

3.3.6 Zero front wheel lean (cross slope (2D) and single 3D systems)

For systems using cross slope or single GNSS operation, remove any front wheel lean.

Operating the machine with the front wheels leaning raises and lowers the front of the goose neck and results in an error in the mainfall measurement. For high accuracy, always operate without wheel lean.

3.3.7 Perform a solenoid calibration

On EH machines, perform a solenoid calibration before performing the valve calibration. An incorrect solenoid calibration will cause poor automatic hydraulic performance.

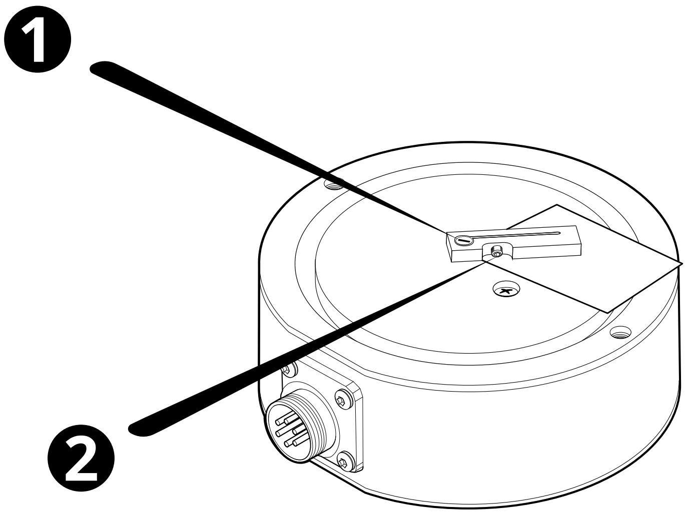
For information on how to perform the solenoid calibration, either see the machine manufacturer documentation or contact the machine dealer.

3.3.8 Align the RS400

On machines with grader blade attachments, align the RS400 rotation sensor to ensure a high accuracy sensor calibration or to enable the sensor calibration to complete. To align the rotation sensor:

1. Position the cutting edge square to the machine.
2. Connect the rotation sensor cable to the rotation sensor.
3. Open the Diagnostics screen in the Web Interface and view the RS400 rotation sensor output. If the output value is:

   a. Within ± 32° of zero - the rotation sensor is calibrated accurately and no adjustment is needed, move on to the next section
   b. Greater than 32° of zero - perform the next step
4. Remove the drive arm from the rotation sensor and then remove the rotation sensor from the mounting plate. Make sure you do not move the sensor arm that is under the sensor.
5. Turn the sensor upside down, making sure to not move the sensor arm. Slide a thick card in between the sensor arm and the sensor base to create space between the two (❶). Hold the sensor arm in place against the card.
6. Use an allen key or torx head screw to loosen the allen screw on the side of the sensor arm (❷). Then use a screw driver to rotate the sensor shaft until the sensor output is within ± 32° of zero on the Diagnostics screen.



7. Tighten the allen screw on the sensor arm. Confirm that the sensor output remains within ± 32°.
8. Remove the card from between the sensor arm and base and slide the sensor arm to check it does not drag on the base.
9. Reattach the rotation sensor to the mounting plate and reattach the drive arm.
10. Confirm that the sensor output remains within ± 32°.
11. Check that all bolts are tightened firmly.
12. Confirm that the sensor output remains within ± 32°.

3.4 Calibration

3.4.1 Calibration process

You can calibrate the sensors in any order:

- Body sensor
- Blade sensor
- Blade rotation sensor

The Web Interface describes the steps for each sequence.

3.4.2 Leave machine stationary

During the sensor calibration process, do not move the machine unless requested by the instructions.

3.5 Mark measure-up points

3.5.1 Best practices

NOTE – Position targets accurately. Your measurements cannot be more accurate than your target positions.

- *Clean* all measurement and control points.
- *Mark* all measurement and control points with a target.
- *Label* all measurement and control points with the point ID. Write the ID beside the target so you can see it through the instrument.

TIP – Use a USB flash drive to transfer measured points from the total station onto the system.

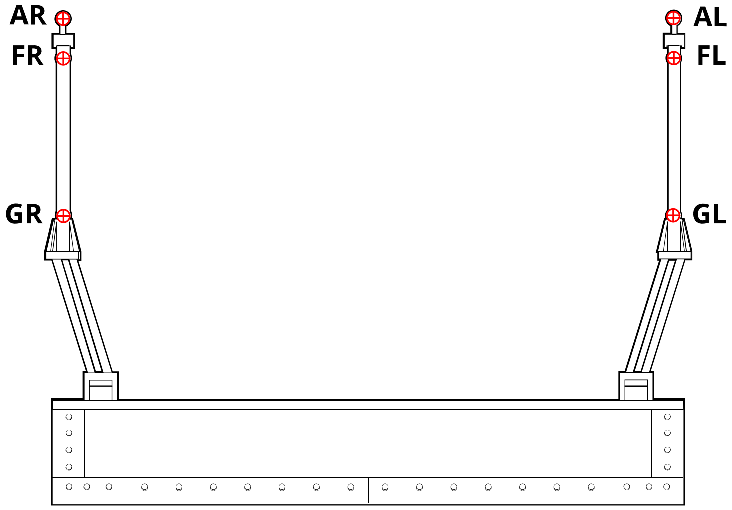
To use this method:

- To enable checking, before you begin the measure-up, make sure that the data collector is set to use the same measurement units and the same coordinate order as set in the Web Interface.
- Name the measured points as they are named in the Measure-up section of the Install Assistant.

For more information, see Using a USB flash drive to transfer measure-up data in .csv format.

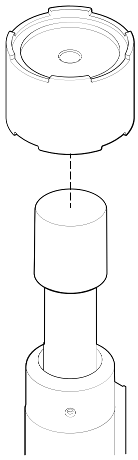
Follow the instructions on the screen to complete the measure-up of the blade, cutting edge, and blade tips.

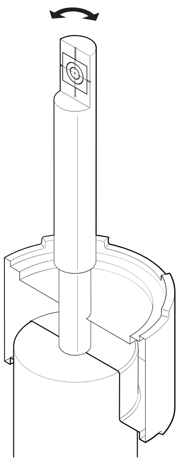
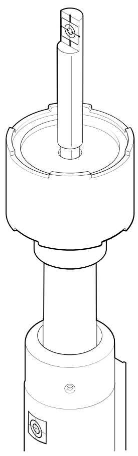
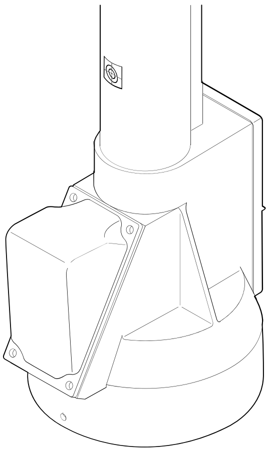
Mark blade-mounted mast measurement points (AL, FL, GL, AR, FR, GR)



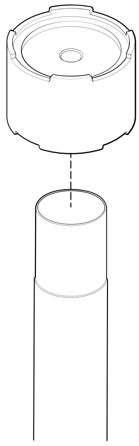
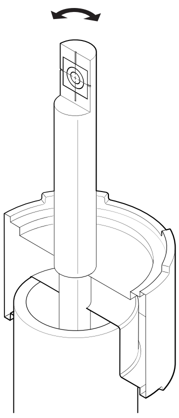
You must use a measure-up target kit (P/N 105568) to measure the AR and AL mast points.

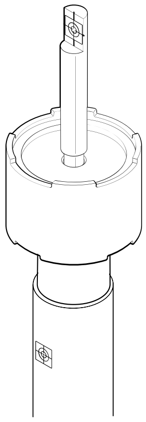
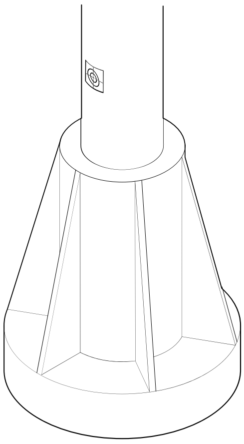
Electric mast

Steps	Description
	Mount the base of the measure-up target on top of the mast so that the side with the narrower diameter hole is on top. **NOTE –** Ensure that the plastic cap that covers the top of the mast is present, intact, and fitted securely to it.

Steps	Description
	Insert the target rod into the base. Ensure that the bottom of the rod rests on top of the plastic cap that covers the top of the electric mast. **NOTE –** If the cap is broken and the base of the rod extends into the hollow space inside the mast, do not proceed with the measure-up as guidance will be inaccurate. Replace the cap before proceeding. Rotate the rod so that the target faces the expected position of the instrument.
	Attach an acrylic target (FR) to the top of the mast and an acrylic target (GR) to the bottom of the mast. Place FR and GR on the surface of the mast such that a line between FR and GR is parallel to the centerline of the mast.
	

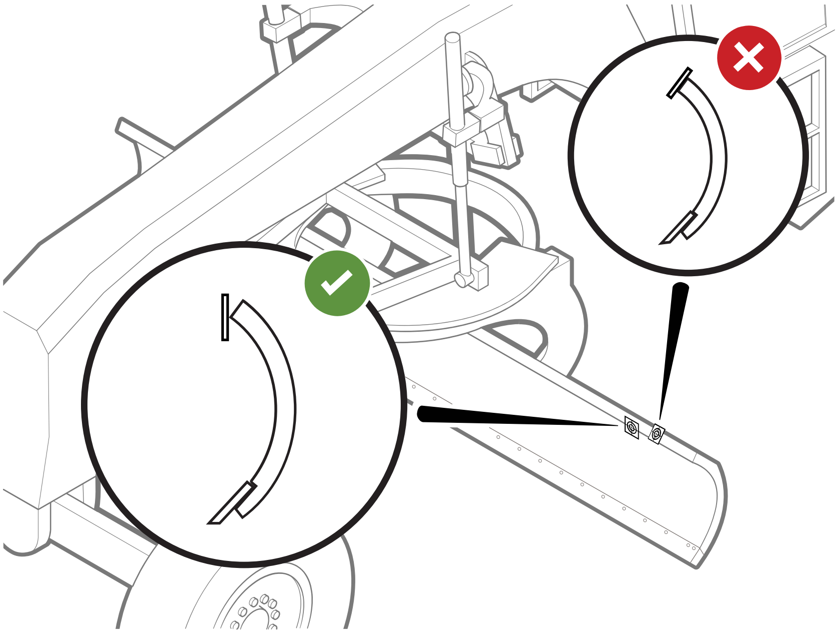
Fixed mast

Steps	Description
	Mount the base of the measure-up target on top of the mast so that the side with the narrower diameter hole is on top.
	Insert the target rod into the base. Ensure that the rod rests on the base of the measure-up target and the end of the rod extends into the hollow space inside the mast. Rotate the rod so that the target faces the expected position of the instrument.

Steps	Description
 	Attach an acrylic target (FR) to the top of the mast and an acrylic target (GR) to the bottom of the mast. Place FR and GR on the surface of the mast such that a line between FR and GR is parallel to the centerline of the mast.

Blade height target placement

When you attach an acrylic target to the top of the blade to measure position E, ensure it is standing vertically so the total station can measure it accurately. If the target lies flat on the top edge of the blade, the angle of the target may cause an inaccurate measurement.



Cab and body mount: mark lift ram ball joint measurement points (IL and IR)

If you are installing a mastless GNSS system or a circle sideshift kit, you must capture points IL and IR at the base of the lift ram ball joints.

If the base of each ball joint is occupied by an automatic grease nipple, you will be unable to place the acrylic target flush on the center of the ball joint. To obtain these points, do one of the following:

- Place the target on top of the grease nipple.
- Offset the target a set distance above the position and then add a target height to the total station measure-up. Ensure the offset is the same amount for both the left and right ball joints.

3.6 Using a USB flash drive to transfer measure-up data in .csv format

You can export comma separated value (.csv) files from most common survey data collectors onto a USB flash drive. You can edit .csv files using a text editor, or by opening the .csv file in a product such as Microsoft® Excel.

> **TIP –** For technicians working in regions where it is common for .csv files to use semicolons to separate fields, and commas as decimal fraction markers, then the Web Interface .csv import tool is also able to import these .csv files.

When formatted correctly, as described below, the .csv file can be imported into the Web Interface from the USB flash drive, reducing the amount of manual data entry required during machine commissioning.

3.6.1 Prerequisites

1. The measurements were made using a data collector running software capable of exporting .csv files.
2. The measurement units configured in the data collector are the same as the measurement units used in the Measure-up section of the Install Assistant.
3. The measured points are named as shown in the Measure-up section of the Install Assistant.

3.6.2 Exporting the machine measurements as a .csv file

On the data collector used to make the machine measurements, export the measured points to a USB flash drive in the usual way. If you are using Trimble SCS900 or Trimble Siteworks software on your data collector, make sure you select Include QA Data in the Export dialog, and select Memory Card as the destination.

3.6.3 If required, edit the .csv file

The final .csv file should resemble the example below when it is opened in a spreadsheet editor. This example shows a typical set of measurements.

The column headings must be:

- In English.
- Spelled as shown.
- Capitalized as shown.

> **NOTE –** The order of the columns and point measurements is not important. You can change the coordinate order for your project to suit the region, and you can measure the points in any order.

If you are editing the .csv file in a text editor, then the contents can look like:

```
Point Name,Northing,Easting,Elevation
```

If you are editing a semicolon separated .csv file in a text editor, then the contents can look like:

```
Point Name;Northing;Easting;Elevation
```

When you save the edited file, you must save it as a .csv file.

3.6.4 Import the .csv file into the Web Interface

1. Insert the USB flash drive into the device running the Web Interface.
2. Proceed through the Web Interface's Install Assistant > Measure-up section as usual until you reach the .csv file upload screen.

   > **TIP –** If you choose the Skip to Entry option when it is offered, you will still have an opportunity to upload your .csv file.

3. In the .csv file upload screen, browse to the required file.

4. Select the Upload button. The file is uploaded and the measured points in the .csv file are copied into the Web Interface.
5. Proceed through the subsequent screens and confirm that the point data has been correctly imported.

> **NOTE –** After measuring the points, the machine ***must not be moved*** until the .csv file import completes, and all the screens in the Measure-up section have been completed.

3.7 Measure drawbar angle

When taking the drawbar angle measurement within the Measure-up section, ensure that the level is:

- Parallel to the machine centerline.
- Flush against the underside of the circle.
- Placed on smooth surfaces only. Avoid any weld beads or imperfections.

> **NOTE –**Placing the level on a weld could result in inaccurate guidance.

4

Grader measure-up and sensor calibration validation

In this chapter:

4.1 Overview of grader measure-up and sensor calibration validation

4.1.1 Overview

The sensors used in the system hold their calibrations very well and are compensated for temperature. You should not recalibrate them unless you have verified a problem. The blade and body sensors are calibrated as part of the Measure-up section in the Install Assistant.

The validation procedures check the accuracy of:

- 2D cross slope machines:
  - Driving a level pass in both directions and ensuring the passes match.
  - Driving a sloping pass in both directions and ensuring the passes match.
- 3D GNSS machines:
  - Stationary blade guidance on a level surface
  - Stationary blade forward roll on a level surface
  - Stationary blade back roll on a level surface
  - Stationary blade tilt to the left
  - Stationary blade tilt to the right

For 2D machines, perform the level surface test before the sloping surface test.

If the required accuracy is not achieved, the procedures describe diagnostic methods for identifying issues.

4.1.2 Prerequisites

Before starting the machine measure-up and sensor calibration validation:

- Install the system components, and validate the physical installation.
- Complete all the sections in the Web Interface's Configure > Install Assistant screen.
- In the Operator App's Work Settings > Blade screen, verify the cutting edge lengths are correct.
- Know how to set-up and use a total station or a GNSS rover, and data collector.
- Validate the total station or GNSS rover on a known control point.
- If you are using a GNSS rover, make sure both the Earthworks system and the rover are using the same site calibration.
- For validating cross slope (2D) systems, use a site with:
  - A level test surface approximately 80 m (260 ft) long and 5 m (16 ft) wide.
  - A second test surface approximately 80 m (260 ft) long and 5 m (16 ft) wide, with a cross slope of at least 10%.
- For 3D machines, use a site with a level pad.
- Review this entire guide. This ensures that you understand the full validation workflow.

4.1.3 Equipment list

- A total station in good adjustment or a GNSS receiver, with a fully charged battery
- If required, a data collector with a fully charged battery
- A note book and pen
- These instructions

4.2 Validating cross slope (2D) systems

4.2.1 Validating cross slope (2D) accuracy on the level

Check

1. Set the machine in a neutral position:
   - Center the drawbar
   - Remove machine articulation

   For guidance on performing these steps, see Chapter 3, Grader sensor calibration and measure-up best practices in the Web Interface or online.
2. Set a target cross slope: it can be either level or sloping.
3. Perform a first pass with the machine. Ensure the blade is cutting material for the entire pass.
4. Turn the machine around.
5. Return over the first pass. On the return pass, the blade should match the surface from the first pass within 0.5%.

Corrective action

If the return pass does not align with the first pass:

1. Ensure you have completed Step 1 above correctly.
2. Repeat the Sensor Calibration and Measure-up sections of the Web Interface Configure > Install Assistant screen.

4.2.2 Validating cross slope (2D) accuracy on a slope

Check

1. Set the machine in a neutral position:
   - Center the drawbar
   - Remove machine articulation

   For guidance on performing these steps, see Chapter 3, Grader sensor calibration and measure-up best practices in the Web Interface or online.
2. Place the machine on a swath of ground with a cross slope of at least 10%.
3. Set a target cross slope.
4. Perform a first pass with the machine. Ensure the blade is cutting material for the entire pass.
5. Turn the machine around.
6. Return over the first pass. On the return pass, the blade should match the surface from the first pass within 0.5%.

Corrective action

If the return pass does not align with the first pass:

1. Ensure you have completed Step 1 above correctly.
2. Repeat the Sensor Calibration and Measure-up sections of the Web Interface Configure > Install Assistant screen.

4.3 Validating 3D systems

4.3.1 Check stationary blade accuracy

Check

> **NOTE –** For single 3D systems, before repeating step 3 in the following procedure:
>
> - The machine must be moved forward or backward in a straight line for about one machine length. This enables the system to re-estimate the machine heading before the next measurement is made.
> - The blade must be square to the machine.

1. Move the machine onto the level pad.
2. Place the blade on the ground or on blocks with the masts vertical.

3. Survey the positions of the blade tips, and compare the surveyed positions with the reported positions in the Points of Interest part of the Web Interface's Monitor > Machine Positions screen. Note the two sets of positions.
4. Pitch the blade forward 5° and repeat step 3.
5. Pitch the blade 5° back past the vertical position and repeat step 3.
6. Return the masts to vertical.
7. Tilt the blade so the right tip is approximately 1 m (3 feet) above the surface and the left tip is touching the surface, and repeat step 3.
8. Tilt the blade so the left tip is approximately 1 m (3 feet) above the surface and the right tip is touching the surface, and repeat step 3.

Corrective action

Compare the surveyed positions of the blade tips to the reported positions for each scenario. The positions should be within 0.025 m (1 inch) of each other.

If they are not, repeat the Measure-up section of the Web Interface's Configure > Install Assistant screen.

4.4 Save and activate machine files

4.4.1 Save a machine file

The current settings save automatically in the system memory and persist over power cycles. Current settings are overwritten when you activate or restore a saved machine file. Saving the current settings to a machine file makes the settings available for activating or restoring as needed.

> **NOTE –**Users with Operator Plus accounts can access the Machine Settings Files screen to apply machine files. From within the Operator App, tap System Settings > Manage Machine File. To give an operator Operator Plus access, within the Web Interface open Configure > Install Assistant > User Permissions.

Consider when to save a machine file:

- After installing, commissioning and validating guidance accuracy of the system
- Before and after a new valve calibration so you can compare previous settings to the new settings
- After completing and validating a measure-up (delete the previous machine file to avoid using the old settings instead of the new settings)
- After completing a new machine setup. You must complete a new valve air purge followed by a valve calibration and any other required machine calibrations for each machine using additional attachments and sensors. The machine file is then available for activation when the machine setup changes.

To save a machine file:

1. Open the Machine Settings Files page in the Web Interface.
2. Select and then Save Settings.
3. Give the file a unique name and save.

4.4.2 Activate, or restore, a machine file

Activating, or restoring, a machine file is useful for returning the system back to a state that was previously saved.

Most system settings are overwritten with the values from the machine file when that machine file is activated or restored.

After activating a machine file, check the following items to ensure accurate guidance:

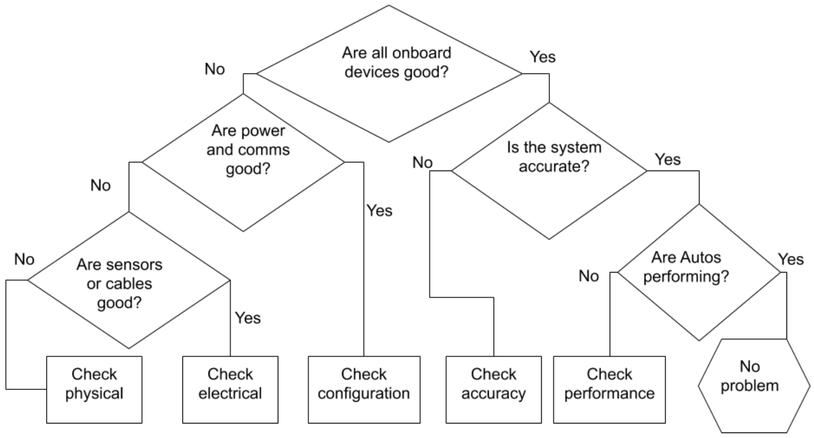
- Cutting edge wear
- Currently loaded project and 2D bench or 3D design settings
- Operational settings such as vertical guidance, focus and overcut protection options
- Items such as Wi-Fi configurations and delayed shutdowns do not reset when activating a machine file.
- System licenses are not affected by activating a machine settings file.

To activate a machine setting file:

1. Open the Web Interface and then Advanced > Machine Settings.
2. Select the file to activate.
3. Select Make Active.

5

Troubleshooting decision tree



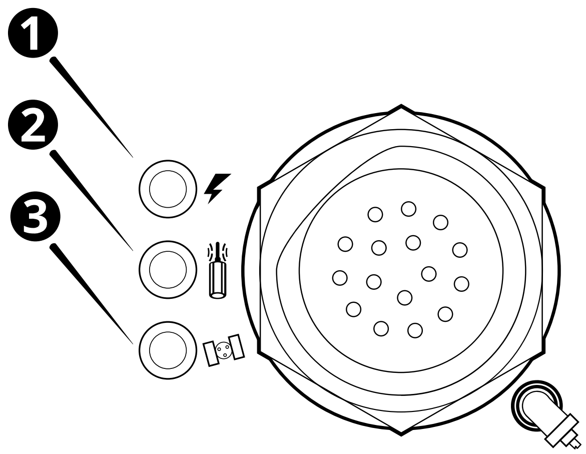
5.1 Check physical

Possible issue	Possible issue and resolution
TD5x0 does not turn on	Cables not connected. Check that the power cable between the TD5x0 and the power source is connected.
	Cable is connected but broken. Check that the power cable between the TD5x0 and the power source is intact and does not have any breaks.

Possible issue	Possible issue and resolution
The right GNSS data link LED is flashing 1 Hz and solid on the left GNSS data link LED and the machine is facing the wrong direction in the guidance views	The left receiver is plugged into the right side cable and the right receiver is plugged into the left side cable. Locate the GNSS splitter cable (P/N 150750-01) and swap the eight pin connectors to the correct sides.
The display shows GNSS Error, No GNSS Position	The GNSS receivers are not connected, check cables and connections or swap the display cable with another cable.
In the Web Interface, when measuring the attachment position, I cannot start the wizard because the GNSS errors are too large or do not have an RTK position.	The GNSS receiver does not receive corrections. Try one of the following: • Ensure you used the correct GNSS measure-up cable (P/N 150775-46). Using incorrect cables or extension cables may increase resistance of the CAN bus. • Note the resistance on the CAN bus and compare to working systems. • Ensure the measure-up cable is attached in the correct location as specified in the instructions on the screen.
On the TD5x0, the wrong action happens when I touch the screen.	Try one of the following: • Be more accurate with your tap gestures. • Activate glove mode, which increases the screen sensitivity. In the TD5x0, open the Settings > Touch > Touch Sensitivity and select Glove.

The MS9xx GNSS receivers have three LEDs next to the harness connector. How these LEDs behave indicates the receiver status, as shown in the following figure.

In a dual GNSS system the left receiver data link should flash at 1 Hz and the right receiver should be solid.



LED	Left	Right
❶ Power	Off – No power On – Power *	Off – No power On – Power *
❷ Data link	Off – No corrections 1 Hz – Corrections †	Off – No corrections On – Corrections received from the left receiver †
❸ Satellite	Off – Not tracking 2 Hz – Tracking fewer than 4 satellites 1 Hz – Tracking more than 4 satellites *	Off – Not tracking 2 Hz – Tracking fewer than 4 satellites 1 Hz – Tracking more than 4 satellites *

* required

† in a dual GNSS system the left receiver data link should flash at 1 Hz and the right receiver should be on (solid)

5.2 Check electrical

5.2.1 TD5x0 display does not turn on

Possible issue	Possible issue and resolution
The display or other devices intermittently powers on and off.	The earth connection may be bad. Test the voltage while the system is off. Then continue to test the voltage and power cycle the whole system to test the voltage under full load.

Possible issue	Possible issue and resolution
When the machine ignition switch is turned on, the display does not turn on.	The boot configuration is not set to turn on when power is applied. Press the black power button on the display. If the display • turns on -consider changing the boot configuration setting in the display system menu. • does not turn on - continue to further tests.

5.2.2 Further tests

1. Check any components with power LEDs (i.e. a GNSS receiver) that the power LED turns on when the ignition switch turns on. Lightbars, if used, should display a short start up pattern when the ignition switch is turned on. If component power LEDs or lightbars:
   a. Turn on, but the TD5x0 does not turn on, the display cabling or the TD5x0 may be faulty.
   b. Does not turn on, the power relay or power cabling to the system may be faulty, perform the next step.
2. Measure the voltage between pins D and E on the circular eight pin display cable connector and pins 1 and 4 on the four pin connector on the display cable. If power is:
   a. Detected – the display may be faulty, replace with a working display.
   b. Not detected – perform the next step.
3. Check the power cable. For machines with a:
   a. Generic power cable (P/N 150411-04) – Disconnect the power cable at the 6 pin Deutsch connector. Measure the voltages between pin 1 (Batt+) and pin 2 (GND) for a constant battery voltage, 12V or 24V. Also measure the voltage between pin 3 (Key switch) and pin 2, which should have the same voltage when the ignition switch turns on. If the expected machine voltage is not detected, check the fuse on the appropriate supply wire. Replace as necessary with the correct fuse (Key switch is ATC 5A Blade type, Batt+ is ATC 15A Blade type). Also check the ground wire connection to the machine chassis – clean or repair if needed. Replace power cable if needed.
   b. Machine-specific power cable – Refer to the wiring schematic for your machine. Trace the power wiring back from the TD5x0 display and machine power cable. Measure voltages at each mating connector pair to identify possible cable faults. Replace the cables and harnesses as required.

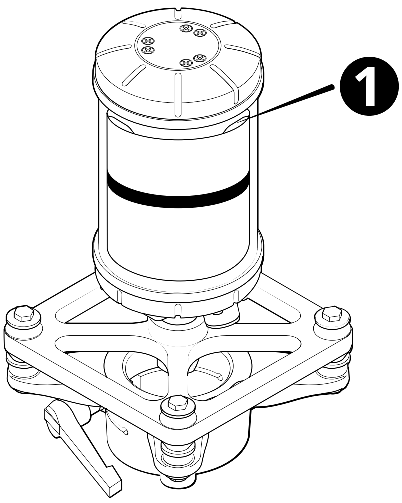
5.2.3 Other electrical issues

Possible issue	Possible issue and resolution
The app is slow on my BYOD	Too many devices may be connected to the Wi-Fi channel, consider using: • a different Wi-Fi channel on the EC520 • a display with a cable connected
The system status tile on the dashboard is red and the Start button is not available. The GNSS power LED light is off.	The GNSS cable is disconnected. Confirm that the GNSS cable is connected.
	No power to the GNSS. See 5.1 Check physical for resolutions.
The operator sees the GNSS ERROR No GNSS Position message.	No power or communication to the GNSS receiver. Check the power and communication connections. No power to the GNSS. See 5.1 Check physical for resolutions.
The LED status indicators on the LR410 laser receiver are not illuminated **NOTE** – The LR410 is unavailable on some machine types.	No power to the LR410. No power to the GNSS. See 5.1 Check physical for resolutions.

5.2.4 MT9xx machine target status indicators troubleshooting

The MT9xx is unavailable on some machine types.

The MT9xx machine target has four status indicators, located at the top of the target (❶). The status indicators show if the unit is receiving power and functioning correctly.



The indicator patterns showing the machine target status are as follows:

Indicator flashing pattern	Meaning
Slow flash (0.1 second on, 0.9 seconds off)	Normal operation
Not illuminated	No power
Fast flash (0.1 second on, 0.1 second off)	Power up (approximately 0.5 seconds), otherwise low battery (<9 VDC)
Blink (3 seconds on, 0.1 second off)	Hardware fault

5.2.5 GS5x0 sensor troubleshooting

If the system reports a GS5x0 sensor error, perform the following troubleshooting steps:

1. On the Web Interface's Onboard Devices screen, select the Recheck button to re-check the error. If it clears, the issue was temporary.
2. Review the error message on the System Status screen:

Error message	Possible issue and resolution
Not Found	The system cannot detect the sensor at all. Possible causes are: – a cable error – incorrect CAN termination – sensor failure If the sensor is not found directly after installation, check that the harness has the correct CAN terminators. If the sensor is not found during operation, check the cables connecting the sensor.
Sensor issue detected	The sensor is communicating, but there is an error. With system power on, use a multimeter to check that the voltage to the sensor is correct: – The voltage between ground and the CAN lines should be 2 to 3 V. – 0 V indicates a short to ground. – Machine V (for example, 24 V) indicates a short to power. In either case, look for a crushed or cut wire. If the error occurred during machine start-up, the cause may be a temporary voltage drop while the engine draws power.

5.3 Check configuration

Symptom	Possible issue and resolution
User preferences are lost after transferring files from a USB flash drive	When transferring files from a USB flash drive to the machine, all files on machine that have the same name as files on the USB flash drive will be overwritten with the USB flash drive files. This includes the user preferences file userdata.pref.xml, which saves the settings; for example, the text items that are configured. To keep the user preferences saved on a machine, either: • Perform a File Transfer of the EarthworksData to the USB flash drive first, then perform a File Transfer from the USB flash drive to the machine OR • Untick the EarthworksData box when transferring files from the USB flash drive to the machine **NOTE –**The userdata.pref.xml file is stored at ProjectLibrary > EarthworksData > [Machine Name Folder]
Machine parts are not seen in the Operator Interface and the Web Interface	Parts may have power supply issues. See Chapter 5.2, Check electrical
The license tile on the dashboard is red	A license is missing: Check that the required licenses for the current machine configuration have been purchased and are installed on the EC520, TD5x0 and/or GNSS receivers.
Licenses installed on the display do not appear in the Licenses screen in the Web Interface	Open the Operator Interface and allow it to upgrade itself to the same version as the server.
License will not install	Wrong license for the device. Compare the serial number on the license file name against the device serial number, they must match.
	The license file is corrupt. Go to the Trimble store and download the file again and install again.

Symptom	Possible issue and resolution
The required licenses have been purchased and installed but the above troubleshooting steps have not resolved the issue.	Isolate and restart the machine, the license should appear.
	Contact support.
All .sg6 files are invalid or corrupt after upgrading licenses	The licenses on the TD5x0 do not automatically upgrade. Open the Operator Interface to start the upgrade.
My Operator App reports a missing device.	The EC520 and device did not communicate correctly, or the installer did not configure the device correctly. To resolve: 1. Open the onboard software. 2. Open Dashboard > System Status and tap Recheck. 3. Check the status indicators beside each device. If the indicator is a. Green - resolved b. Red - power cycle the machine and Operator App If power cycling does not resolve then check the device and harness for damage. If no damage, see 5.2 Check electrical .
My device is not listed on the Onboard Devices list	The missing device is disconnected. Connect the missing device.
A low accuracy error message appears on the display and there is no cut/fill guidance. The left GNSS data link LED is off.	No correction messages are being received from the base station. Check that the correct correction source is selected in the Operator App.
Depth and slope mode only available on my excavator. I cannot select design mode.	No 3D Design License. Contact your dealer. Confirm GNSS is on in the System Status tile. Turn on if off. Power cycle the machine if you have licenses on the display.

Symptom	Possible issue and resolution
Single GNSS on Cat Next Gen excavator is not working correctly.	Single GNSS on Cat Next Gen excavators requires information from the machine system to function. Faults on the machine side may prevent Single GNSS on Cat Next Gen excavators from functioning. If a fault occurs, it will show on the machine display, not the TD5x0. In particular, you must address faults from the machine swing sensor (4326-XX) and machine travel controls (5332-XX or 5352-XX) for correct operation.
In the Web Interface, when measuring the attachment position, I cannot start the wizard because the GNSS errors are too large or do not have an RTK position.	The GNSS receiver does not receive corrections. Try one of the following: • Ensure you used the correct GNSS measure-up cable (P/N 150775-46). Using incorrect cables or extension cables may increase resistance of the CAN bus. • Note the resistance on the CAN bus and compare to working systems. • Ensure the measure-up cable is attached in the correct location as specified in the instructions on the screen.
Cannot synchronize with Connected Community	Check the signal strength icon in the status bar. The system may have no connection, a poor or intermittent connection, poor cell reception, or poor Wi-Fi.
	Check the Web Interface for Connected Community settings and any error messages, and check that Connected Community is enabled: Network > Cloud Services
	Check the Web Interface for gateway device messages: Network > Gateway
	Confirm the data APN setting is set on the SIM card.
	Close the Operator App and load a web page on the Android screen.
In the Web Interface, I can open Configure > Install Assistant > Mast Measure-up, but I cannot start the wizard because the GNSS errors are too large or do not have an RTK position	Verify that the machine's GNSS positioning is loaded and working. The system may not be receiving corrections from the base station. To resolve: 1. Open the Web Interface, exit the Install Assistant > Mast Measure-up screen, open Network > GNSS Correction Source, and check that the correct correction source is selected or add a Correction Source. 2. Complete Configure > Install Assistant > Mast Measure-up.

<table>
<tr><th>Symptom</th><th>Possible issue and resolution</th></tr>
<tr><td>My user name is not on the list on the log in screen.</td><td rowspan="2">To resolve:
1. In the Web Interface, open Advanced > User Permission, and add the names of users who need permission to open the onboard software.
2. Select Operator for the needed user and save.
TIP – Installers - create operator and administrator permissions immediately after completing the Configure section for each machine.</td></tr>
<tr><td>In the TD5x0, I touch the icon to open the application. The log in screen appears, but I get the error message, "No operator configured for this machine"</td></tr>
<tr><td>The precisions on the Web Interface change from green to red to green during a 3D mast measure-up using GNSS receivers.</td><td>No issue. Wait for the measurements to finish recording, then continue with the measure-up.</td></tr>
<tr><td>The operator sees the "Radio does not support UTS" message</td><td>The on-machine data radio does not support the data transfer protocol used by the UTS instrument. Set the on-machine data radio to the 2400 MHz radio band.</td></tr>
</table>

To check:

- the attachment measure-up process, open Configure > Attachments.
- the measure-up process, open Configure > Install Assistant > Measure-up.
- the 3D sensors measure-up process, first check the accuracy of the body sensor calibration process by checking the machine pitch and roll text items in the software of the machine against a calibrated digital level. Next, confirm you haven't changed the GNSS receiver mounting locations. Finally, redo Configure > Install Assistant > Mast Measure-up.

LED	Cause	Resolution
The green LED is off.	Suspect a power supply problem.	Check the power supply to the valve module. Use the connection diagram and schematics to check that the valve module is connected correctly.

LED	Cause	Resolution
The green LED is flashing.	Suspect incorrect termination.	Check all terminators are present and connected. The resistance between CAN hi and CAN lo must be about 60 Ω. Otherwise, suspect CAN driver issues.
The green LED is on continuously.	Suspect incorrect CAN bus connection.	Check all connections. Check that you have used the correct cables.
The red LED is either flashing or on continuously.	Suspect a valve module fault.	Replace the valve module.

5.3.1 Check compatibility of GNSS receivers

The compatibility of receiver pairs depends on the machine type and the firmware that is installed on each receiver.

Blade-mounted receivers - permitted combinations

Receiver	Compatible receiver combination	Firmware requirement
MS992	MS992	firmware must match
MS995	MS995	firmware must match
MS996	MS996	firmware must match

NOTE –The receiver firmware must match or be greater than the minimum required version.

5.3.2 License troubleshooting

Symptom	Possible issue and resolution
The license tile on the dashboard is red	A license is missing: Check that the required licenses for the current machine configuration have been purchased and are installed on the EC520, TD5x0 and/or GNSS receivers.
Licenses installed on the display do not appear in the Licenses screen in the Web Interface	Open the Operator Interface and allow it to upgrade itself to the same version as the server.

Symptom	Possible issue and resolution
License will not install	Wrong license for the device. Compare the serial number on the license file name against the device serial number, they must match.
	The license file is corrupt. Go to the Trimble store and download the file again and install again.
The required licenses have been purchased and installed but the above troubleshooting steps have not resolved the issue.	Isolate and restart the machine, the license should appear.
	Contact support.
All .sg6 files are invalid or corrupt after upgrading licenses	The licenses on the TD5x0 do not automatically upgrade. Open the Operator Interface to start the upgrade.

5.4 Check accuracy

Symptom	Possible issue and resolution
The display shows GNSS Error, Autonomous GNSS	Corrections from the base station are not being received. Check the Correction Source selection is correct, if the problem persists check that the base station is still broadcasting corrections. If you are using an internet delivered correction source, check that you have an internet connection and either cell or Wi-Fi in the area you are working in.

Symptom	Possible issue and resolution
While operating the machine, an orange alert banner appears with the message GNSS precision outside the set tolerance. I looked at the text bar and the cut/fill values are not there. The left GNSS data link LED is off.	The data coming from the GNSS receiver about GNSS Precision Tolerance is outside the range set in the software. This could be due to loss of corrections from the base station or due to obstructions, such as trees, buildings, etc. Possible resolutions: 1. In the Operator App, open System Settings > GNSS Precision Tolerance, and check the H and V precision values and confirm the appropriate mode is selected for the task. 2. If you are using a radio correction source, consider using a repeater to improve radio signal strength. 3. If you are using an internet-delivered correction source (VRS, IBSS, NTRIP) check the signal strength icon in the status bar. The system may have no connection, a poor or intermittent connection, poor cell reception, or poor Wi-Fi. 4. Ensure the GNSS receivers have a clear view to the sky. 5. The radio or base station may be turned off. Confirm that the base station is outputting corrections by confirming that other GNSS devices on site are receiving corrections. 6. Confirm that the data link LED is flashing and move the machine: – Away from trees, buildings, or other tall obstacles. – Closer to the site base station.
After I finish configuring my machine, in the Web Interface there are no values in Monitor > Machine Positions.	You have not loaded a Project so the software is missing the required coordinate system for populating these fields. To resolve: 1. In the Operator App, create or select a project in Dashboard > Job Setup. Loading a project loads a coordinate system.

Symptom	Possible issue and resolution
The operator needs to confirm connection strength for TLL/VSS/NTrip	1. Open the Operator App and go to System Settings > System Status 2. Press and hold GNSS Receiver - Left 3. Wait for the status screen to open 4. Confirm: a. Integrity ≥ 90% b. Latency ≤ 2 sec
My design does not appear on the TD5x0 after transferring files via a USB flash drive	Confirm that the USB flash drive is fully plugged in with the light flashing. Some USB flash drives have casings that are too wide for the TD5x0 USB flash drive port. Confirm that the file is on the USB flash drive by opening the USB flash drive on a computer. **NOTE –**To ensure the file structure on the USB flash drive is optimally set up for the system, insert a blank USB flash drive into the TD5x0, open Trimble> System Settings > File Transfer.
The operator sees the GNSS ERROR No GNSS Position message.	No valid site calibration or ACS (Automatic Coordinate System) loaded.
Machine is now inaccurate after restarting the machine but was accurate before machine shut down	The GS520 grade sensors must be at operating temperature. In colder temperatures, warm up the Earthworks system for at least 10 minutes after turning on power.
	The GNSS is connected to the wrong base station or the base station may have been moved.

To check:

- the attachment measure-up process, open Configure > Attachments.
- the measure-up process, open Configure > Install Assistant > Measure-up.
- the 3D sensors measure-up process, first check the accuracy of the body sensor calibration process by checking the machine pitch and roll text items in the software of the machine against a calibrated digital level. Next, confirm you haven't changed the GNSS receiver mounting locations. Finally, redo Configure > Install Assistant > Mast Measure-up.

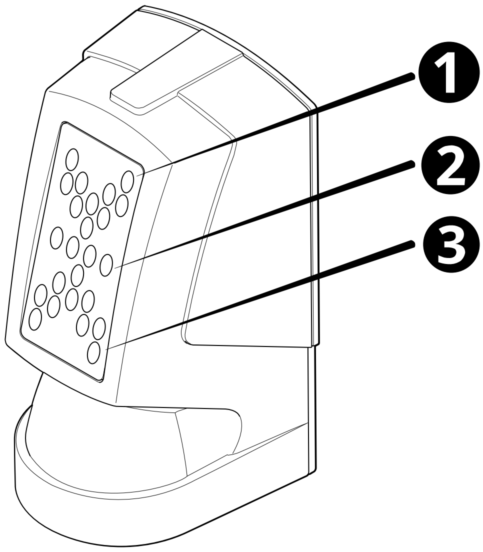
5.4.1 2D-Sensor troubleshooting

The ST400 and LR410 are unavailable on some machine types.

Check ST400 status indicators

The ST400 sonic tracer has three sets of LED status indicators:

- Over-grade (❶)
- On-grade (❷)
- Under-grade (❸)

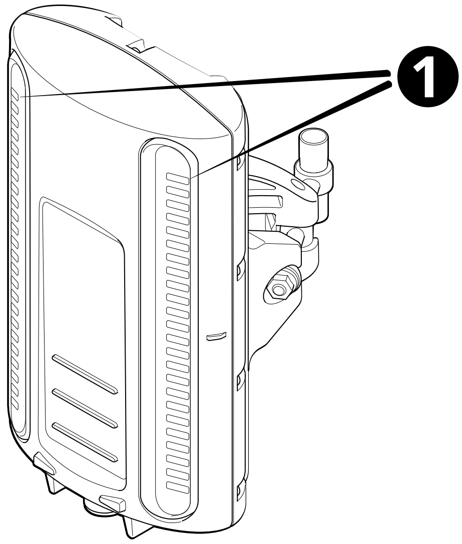


LED pattern	Meaning
Down arrows flashing alternately	Above the sonic gate *. Lower the sonic tracer so that it is within ± 70 mm (2.8 inches) of the benched elevation.
Up arrows flashing alternately	Below the sonic gate *. Raise the sonic tracer so that it is within ± 70 mm (2.8 inches) of the benched elevation.
Outer up and down arrows flashing alternately	No echo detected. The sonic tracer may not be benched.

* The sonic gate is the region within which the sonic tracer must be operated. This is ±70 mm (2.8 inches) of the benched elevation.

Check LR410 status indicators

The LR410 laser receiver has two sets of LED status indicators (❶).



LED pattern	Meaning
Not illuminated	No power
Slow flash (0.4 seconds on, 1 second off)	No strike
Flash (0.2 seconds on, 0.2 seconds off)	Strike detected above center of the receiver
Fast Flash (0.1 second on, 0.1 second off)	Strike detected below center of the receiver
On continuously	Strike detected at the center of the receiver

5.4.2 2D system troubleshooting

Symptom	Possible issue and resolution
Guidance lost due to laser interference	An LED or similar light source on the body of the machine is interfering with the operation of the laser receiver. Raise the laser transmitter and receivers, if possible, until guidance is restored.
When benching a laser receiver, the operator gets the error message "No laser strike detected"	Too much variation in the output of the machine's angle sensors. Ensure the machine is stationary and then bench it.

Symptom	Possible issue and resolution
When benching a laser receiver, the operator gets the error message "Not enough laser strikes"	Check transmitter line of sight and stability. On sites with multiple laser transmitters, check that the receiver is not detecting a second laser strike.
When benching a laser receiver, the operator gets the error message "Too much blade movement"	Too much variation in the output of the machine's angle sensors. Ensure the machine is stationary and then bench it.
When benching a sonic tracer, the reference surface is out of range	The sonic tracer may be out of range. The operating ranges for an ST400 sonic tracer are: Stringline: 200 mm – 900 mm (8 inches – 36 inches) Surface: 200 mm – 1300 mm (8 inches – 51 inches)

5.5 Check performance

To check the machine performance, refer to the tests in the *validation guide*.

Refer to the Autos Performance troubleshooting decision tree available on Partners for other performance checks to consider.

5.6 System errors

This section describes some system errors that may occur.

The Operator App shows simplified error and warning messages.

For detailed error and warning messages, see the Web Interface's Monitor > Program Log screen.

5.6.1 GNSS receiver errors

This section describes possible errors reported by a GNSS receiver.

To view any active GNSS errors, select the Monitor > GNSS Details screen in the Web Interface. Active errors appear at the top of the screen.

Error Code	Meaning	Action
1	Not enough satellites to do any kind of fix.	Check for obstructions between the antenna and the satellites.
2	PDOP too high to do autonomous fix.	Check for obstructions between the antenna and the satellites.

Error Code	Meaning	Action
3–63	Conditions too poor for any positioning.	Check for obstructions between the antenna and the satellites.
64	Wait a few seconds.	The issue may resolve.
65	No base station data available.	Check the radio configuration.
66	Base station coordinates unknown.	Check the radio configuration.
67	Not enough satellites to do differential fix.	Check for obstructions between the antenna and the satellites.
68	Valid reference satellites could not be found.	–
69	Base station is not tracking the same kind of satellite code that rover is.	Check the base configuration.
70	PDOP is too high to do differential fix.	Check for obstructions between the antenna and the satellites.
71–72	Internal error or misconfigured RTK engine.	Contact your support channel.
73	Wait a few seconds.	The issue may resolve.
74–75	Too many satellites blocked or geometry too poor for a precise position.	Check for obstructions between the antenna and the satellites.
76	Internal error or misconfigured RTK engine.	Contact your support channel.
77	Exceeded baseline limit set in option blocks.	–
78	No initial RTK rover position.	The issue may resolve.
79	Base station tracking is weak.	–
80	Tracking is too weak or there is too much multipath for a precise position.	Check for obstructions between the antenna and the satellites.
81	Missing multi-band reference information.	Check the base configuration.
82	Missing multi-band rover information.	The issue may resolve.

Error Code	Meaning	Action
83	Contact your support channel to review receiver options.	Contact your support channel to review receiver options.
84–127	Conditions too poor for differential positions.	Check for obstructions between the antenna and the satellites.
128	Could not find sufficient L2 phase.	Check for obstructions between the antenna and the satellites.
129	RMS value for RTK fixed position is too high.	Check for obstructions between the antenna and the satellites.
130	Wait a few seconds.	The issue may resolve.
131	Internal error or misconfigured RTK engine.	Contact your support channel.
132	Insufficient satellites to carry out search.	Check for obstructions between the antenna and the satellites.
133	Internal error or misconfigured RTK engine.	Contact your support channel.
134–135	Too many satellites blocked or geometry too poor for a precise position.	Check for obstructions between the antenna and the satellites.
136	Propagation time exceeded limit.	Check for obstructions between the antenna and the satellites.
137	DOP of search satellites was too high.	Check for obstructions between the antenna and the satellites.
138	Internal error or misconfigured RTK engine.	Contact your support channel.
139	Reference station is too far away to initialize.	–
140–142	Too many satellites blocked or geometry too poor for a precise position.	Check for obstructions between the antenna and the satellites.
143	Internal error or misconfigured RTK engine.	Contact your support channel.
144	Contact your support channel to review receiver options.	–

Error Code	Meaning	Action
145–150	Tracking is too weak or too much multipath for a precise position.	Check for obstructions between the antenna and the satellites.
151–254	Conditions too poor for precise positions.	Check for obstructions between the antenna and the satellites.

5.6.2 WorksOS errors

This section describes possible errors encountered when using the WorksOS online service.

Error Code	Meaning	Action
3047 3100	Ground surface download is not configured correctly: unable to identify device.	Check that your device is registered.
3102	Ground surface download is not configured correctly: device account issue.	The system cannot locate the WorksOS account for the device.
3103		There is more than one active WorksOS account for the device.
3048 3105	Ground surface download is not configured correctly: no office project found for this device.	Check with your office data manager that the device is assigned to the relevant WorksOS project.
-5 3019 3020 3037 3049 Others	Ground surface download error: contact support.	There was an unexpected issue. Contact support for help.
3017 3124 -3	Unable to connect to server. The system will retry shortly. Contact support if this does not resolve.	The system is unable to connect to the server. The system will retry. Check the internet connection. If the system continues to report an error, contact support.
–	Cannot contact server. Ensure your system is connected to the internet.	Check the internet connection.

Error Code	Meaning	Action
–	Download failed: system will retry shortly.	The download failed or was interrupted. The system will retry after a brief pause.
–	There is no office project at this location. Contact your office data manager.	You are outside of any WorksOS project boundary. Check with your office data manager that the WorksOS project for your location is available.
3021	System error. Contact support.	The system is using UTS navigation but there is no .cal file in the project. Add a .cal file so the system can identify the correct project in WorksOS.

5.6.3 License errors

The Status text at the top of the Licenses screen reports any license errors. Possible errors are:

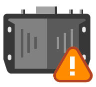
Error Code	Meaning	Action
Network error 6	There is a DNS error.	Check your internet connection. If the issue persists, contact support.
Network error 28 / -1	There is a timeout communicating with the server.	
Other network errors	There is a network error.	
Install error 5	An install is already in progress.	Wait 30 seconds and retry. If the issue persists, restart the system.
Install error 6	The device is not found.	Check the device connection and license service state. If the issue persists, contact support.
Install error 7 / 11 /16 / 17	The display's time is incorrect.	Correct the display's time.
Install error 10	The device's operating system does not support cloud licenses.	Update the device's operating system to a version that supports cloud licensing.
Install error 14	The uploaded license is an option key, but the device requires a cloud license.	Obtain a cloud license.

Error Code	Meaning	Action
Other install errors	The installation failed.	Retry the installation. If the issue persists, contact support.
HTTP error	There was an unexpected issue with proxy or network settings.	Retry later. If the issue persists, contact support.
All other errors	An error has occurred.	

5.6.4 Device errors

To view the status of hardware devices in the system, see the Web Interface's Monitor > Onboard Devices screen.

Each device connected to the system has an icon that shows the current status of the device:



The potential states are:

Symbol	Detail
	The device is functioning with data flowing correctly.
	The device is connected but may be in one of the following states: • Unreliable data • Configuring • Configuration failed • Firmware update required • Not required in the current machine configuration
	• Faulty: The sensor may have a non-recoverable fault. • Not Found: The system cannot detect the device. • Issue: There is a problem with the device. You can probably resolve it without replacement.

For more information, click the expand arrow of a device.

GS5x0 status descriptions

Field	Detail	
Device Connection	Shows whether the sensor is connected over CAN. One of:	
		The sensor is correctly connected.
		**None**: The sensor is required but not connected over CAN.
Device Status	Shows whether the sensor hardware is working correctly. One of:	
		The sensor is working correctly.
		**None**: The sensor is required but not connected over CAN.
		Low Voltage: The sensor is reporting low voltage.
		Replace Sensor: The sensor is reporting a non-recoverable error.
Firmware	Shows whether the sensor's firmware meets the minimum requirement:	
		The firmware meets the minimum required version.
		**Update Required**: Firmware does not meet the minimum required version.

5.6.5 Design validation

The system supports designs in LandXML, .vcl or .dsz format. When a design is added to the system, the system validates the file to ensure it is valid. You cannot use the file until validation is complete.

When a design is queued for validation, the system shows the validating icon .

When the system is validating the design:

- The system displays the validating icon  and a time on the Job Setup screen beside the design's name in the Design File menu.

- The time is an estimate of how long the validation will take.
- A warning message displays:

On-screen Message	Description
File > Validation in progress...	Wait for the system to complete the validation process.

> **NOTE –** If you load a design, the system will pause validation of other designs.

If a design was received automatically from WorksManager, the error or warning is also recorded in the WorksManager system.

Validation timing

The time that the system takes to validate your designs varies depending on the complexity of the design:

- In the Design File drop-down, the system shows a countdown of the estimated time remaining for the design that is being validated.
- To validate a specific design ahead of other designs, select it in the Job Setup screen.

> **NOTE –** If you introduce a project that contains many complex designs, the time to validate them may be significant.

Troubleshooting designs

Files within a project can fail validation for a number of reasons. This section describes some causes. Validation occurs for several file types within a project (not just designs).

> **TIP –** To correct a design that fails validation, you must adjust the design in the office software that generated it (such as Trimble Business Center). Therefore, the information in this section may be most relevant to site engineers or data managers.

> **TIP –** Highly complex designs can fail validation because they require too many system resources. Smaller designs (for example, less than 20 MB) are less likely to cause problems.

Validation troubleshooting

You can receive 2 types of validation message:

- A red cross : the design cannot be used. The file fault will need correcting in the design tool.
- An orange exclamation mark : the design can be loaded, but there is a warning. It may not display fully or provide guidance as expected.

The Operator App provides basic guidance on why a design fails validation. This section describes how to get more detailed information.

When the system validates a design, it creates a text file called *ProjectDiagnostics.txt* in each project's folder that records details of the validation. The contents of this file can help you to locate the cause of the failure and correct it.

> **NOTE –** The file is only generated for new validations, so it won't exist for designs validated on earlier versions of the software.

To obtain the file:

1. Use the File Transfer screen to export the project to a USB drive.
2. On a PC, navigate to the *ProjectDiagnostics.txt* file in the project.
3. View the file in Notepad and scroll to the section for the design that failed validation.
4. Look for the following strings:
   - `Validation State`
   - `Validation Warnings`

   Here is an example from a design that failed validation. *ProjectDiagnostics.txt* states:
   - `Validation State: Invalid`
   - `Validation.Warnings.0: Linework extends outside design boundary (400x400km)`

   This design was invalid because a line element extends beyond the design boundary. There may be more than one reason why a design fails validation. The validation file records the first 5 reasons for failure.

For designs that are usable but generate an orange warning because an element is missing, you can also find detailed validation information in the Web Interface's Monitor > Program Log screen. Look for warnings relating to <NavigationProcessorComponent>. For example:

WARNING - <NavigationProcessorComponent> Validation warning: Linework extends outside design boundary (400x400km)

> **NOTE –** The design may display with expected elements missing. If the element you require is not present, the file issue will need correcting in the design tool.

Issues that stop the system from loading the design include:

On-screen Message	Description
Issue > File Missing > Check File Structure	You cannot load the design because the system could not find the file. This issue can also occur if the file is present but is 0 kB.
Issue > Unrecognized Format > Check Design	You cannot load the design because the system does not recognize the file format.

On-screen Message	Description
Issue > Invalid File > Check Design	You cannot load the design because the file is invalid. This may be because: • The file does not conform to the design schema. (This condition does not apply to LandXML.) • Units of measure are not defined in the file. • The units that are defined are unsupported. • The maximum number of line elements is exceeded.
Issue > Invalid File > File Size Too Large > Check Design	You cannot load the design because the file's size is too large: • For .vcl files, the limit is 150 MB. • For LandXML files, the limit is 50 MB. • For .dsz files, there is no size limit.
Issue > Surface Too Complex > Check Design	There are too many triangles in the design's TIN model (more than 65,0000 in one page).
Missing project coordinate system file	The system cannot find a .cal file for this project. Check the project's \OfficeData\ folder.
No Measured Data selected	Select a Measured Data file in Job Setup screen.

Issues where there is a problem with the design, but you can still load it, include:

On-screen Message	Description
Avoidance zones do not work with a 2D positioning source	The selected project contains one or more avoidance zones, but cannot display them because you are using a 2D positioning source.
Design File > Surface Issue Design loaded, but only valid TIN surfaces will show. Check the design file.	The system can load the design, but there is an issue with one or more surfaces: • The surface may be missing an element. • The surface is not a valid TIN model.

On-screen Message	Description
⚠ Design File > Surface Issue Design loaded, but one or more surfaces contain missing or invalid data and may be unusable. Check the design file.	The system can load the design, but there is an issue with one or more surfaces. Causes include: • Missing faces or points • Invalid points • Invalid face • Face refers to non-existent points • System cannot process the surface TIN model
⚠ Design File > Linework Issue Design loaded, but one or more layers contain missing or invalid linework data and may be unusable. Check the design file.	There are many possible causes of this error. Refer to the *ProjectDiagnostics.txt* file for more detail.

On-screen Message	Description
Linework elements unsupported in LandXML • Line is not 2D or 3D • Curve is not 2D or 3D • Curve has inconsistent radius • Curve length does not match provided length • Curve has undefined bearing or arc length • Invalid point list • Point list has incorrect number of values • Line color is unknown format • Point has invalid coordinates • Point has no name • Curve has invalid rotation • Curve has invalid or no center coordinate • Start/end dimensions do not match • Line has wrong contents • Alignment does not have a name • Alignment has no staStart or length • Alignment total length does not match	• Line only has one horizontal element • Error converting string to double • cgPoint contains no value or reference • Invalid spiral radius • Cannot find point in layer • No layer exists • Circular referencing • Invalid point • Cannot find point in lookup table • Point name should be unique • Feature tag has empty label • Feature tag has empty value • Invalid PI dimension • Spiral start and end point is neither 2D nor 3D • Linework contains bad data • Unsupported linework type

<table>
<tr><th>On-screen Message</th><th>Description</th></tr>
<tr><td colspan="2">Linework elements unsupported in VCL:</td></tr>
<tr><td>

Unsupported linestring elements:

Dependent Location Three Point Arc
Dependent Three Point Arc
Dependent Segment Three Point Arc
Dependent PI Arc with elevation
Offset with useSlope
Offset endpoint with negative along
Endpoint with StationOfffset

Unsupported linestring vertical elements:

eStationSlopeToParabola
eStationSlopeFromPI
eStationSlopeFromParabola
eStationSlopeToFromPI
eStationSlopeToFromParabola
eStationSlopeToArc
eStationSlopeFromArc
eStationSlopeToFromArc

Singular best fit arc linestring element
Best fit arc linestring element with the same start and end points
Smoothed polylines
Smooth Curves

</td><td>

RoadIntersectionLinestring
UtilityLine
Hatch
Moss6D
Linestring with no elements
3D spirals
Smart text
Decreasing stationing
Point Coordinate Component
Point Coordinate Type
Vertical alignment segment of type “<type>”
Offsets to alignment spirals
Any entities defined on a cutting plane
Any entities defined using a User Coordinate System
Composite surfaces
Only single point coordinates supported, <number> points found
Only alignments with curve type = “arc” are supported
Only alignments with spiral type = "Clothoid" or "Cubic" or "Cubic Parabola" or "NSW Cubic Parabola" or "Half Sine" are supported

</td></tr>
</table>

Design complexity limits

A design can also fail validation if it contains too many elements. Exceeding the following limits can cause a design to fail validation:

Element	Limit
Too many lines or linework elements (including curves)	65,534 lines in one page.

Element	Limit
Dense surface	Avoid using surfaces where the TIN triangles are too small, e.g. under 0.01 m^2 per triangle or has more than 65,000 triangles in an area of 50 m x 50 m.
Too many points	160,000 points.
Oversized TIN triangles	TIN triangles with 40 km sides. If you require extremely large TIN triangles, use .dsz format.
Avoidance zones	800 segments (or sides) per polygon. The system supports multiple avoidance zone polygons. For best performance: • Try to keep individual polygons below 500 segments. • Break up a large or thin polygon into multiple polygons with similar dimensions.

In addition, use of the following may cause issues within a design:

- Contains a spiral, horizontal cubic, or NSW spiral
- 3D lines of curves
- Complex vertical segment arcs with horizontal spirals
- Ferguson linestring elements
- Excessive text/labels
- Excessive line names
- Excessive layers
- Design triggers many warnings/information notes

5.6.6 TCC (Connected Community) errors

This section describes possible errors encountered when using the TCC online service.

Error Code	Meaning	Action
0	Success: The files synced successfully with TCC.	N/A
2	Canceled: Either the technician canceled the TCC file sync, or an event occurred that canceled the sync (for example, an operator logging into the Operator App).	If you did not cancel the sync, consider whether an event (such as operator login) caused the cancellation. Repeat the sync.

Authentication errors:

Error Code	Meaning	Action
10	Failed login: the login attempt failed because of an account issue	Ensure that you are using the correct login details and that your subscriptions are valid.
11	No credentials	TCC sync failed because the machine file is missing a required machine name or organization name.
12	Machine name not set	

Networking errors:

Error Code	Meaning	Action
20	Server connection failed	Check the internet connection is active and the TCC configuration details are correct.
21	HTTP host not resolved	
22	Network unreachable	Check if the network is blocking data.

Error Code	Meaning	Action
23	HTTP request failed	When the grade control system attempted to communicate with TCC, the online service returned one of the standard HTTP error codes (you can review the meaning of your specific code online). Common codes include: • 400 – Bad request. • 401 – Unauthorized. • 403 – Forbidden. • 404 – Not Found: The requested URL does not exist. • 429 – Too many requests. • 502 – Bad Gateway: When you connect to a website, your request is routed through a string of servers. A server at some point between your system and TCC has rejected the request. The error may resolve at a later point simply because a different set of servers is used to reach TCC. It may also be caused by your system configuration. • 503 – Service unavailable. • 504 – Gateway timeout. • 511 – Network authentication required.

Server errors:

Error Code	Meaning	Action
30	Remote files do not exist	The file that you are attempting to download from TCC does not exist (for example, because it was deleted, making your list of files out of date).
31	Should not continue	The server reported an error.
32	List remote directory	

Error Code	Meaning	Action
33	List file returned multiple files	An error occurred because the system requested details of one file but received details for multiple.
34	Create remote directory	The server reported an error.
35	Cannot uplink zero-sized file	An error occurred because the grade control system tried to upload a zero-sized file to TCC.
36	Rename remote	The server reported an error.
37	Delete remote	
38	Delete remote directory	
39	Delete remote lockfile	
40	Get remote modify time	
41	Parsing JSON	The server sent a response that is invalid.
42	Uplinking chunk to remote	
43	Invalid file space ID from remote	

Local file system errors:

Error Code	Meaning	Action
60	Insufficient space	Free up space on the EC520.

Error Code	Meaning	Action
61	The grade control system failed to rename a file locally.	The local system reported an error.
62	The grade control system failed to delete a local file.	
63	The grade control system failed to create a directory locally.	
64	The grade control system failed to delete a local lock file.	
65	The grade control system failed to open a local file.	
66		
67	The grade control system failed to update the time on a downloaded file.	

Other errors:

Error Code	Meaning	Action
80	The system is already trying to log in.	Wait for a short period.
81	The system is already trying to sync with TCC.	Wait for a short period.
83	Not permitted.	–

6

TD5x0 Android Recovery Mode

In this chapter:

6.1 Recovery Mode

Recovery Mode provides an option to clear a TD5x0 hard drive. This option is useful for resetting a TD5x0 Android password. This option resets the Android device password only and does NOT reset the Trimble® Earthworks password.

6.2 Reset a TD510 or TD520 Android password

To reset the Android password on a TD510 or TD520:

1. Turn on the display. Press and hold the power button to open an options menu.
2. Tap Recovery and then OK on the Recovery Mode box.
3. Wait for the No Command, or dead Android, image to display.
4. Press simultaneously the power and volume up button a few times until the Android Recovery menu opens.
5. Follow the on-screen instructions to select wipe data/factory reset.
6. Wait for the display to reboot and enter a new password.

> **NOTE –**Wipe data/factory reset erases all files, such as design files, from the display's hard drive. All installed apps remain.

6.3 Reset a TD540 Android password

To reset the Android password on a TD540:

1. Turn off the display. Press and hold the power and volume down buttons at the same time until the display turns on.
2. Wait while the device image displays.
3. Press the volume up button for five seconds and release. This makes the menu open.

4. Follow the on-screen instructions to select wipe data/factory reset.

   > **NOTE –** Tapping the screen also scrolls through the menu items.

5. Wait for the display to reboot and enter a new password.

> **NOTE –** Wipe data/factory reset erases all files, such as design files, from the display's hard drive. All installed apps remain.

6.4 Exit the No Command screen

To exit the No Command, or dead Android, screen (before the Recovery Mode menu screen), press the display's power button or leave the display on for two minutes and it will automatically reboot.

6.5 Exit Recovery Mode menu

To exit the Recovery Mode menu, follow the on-screen instructions to select reboot system now.

7

Supported third party tablet installation

In this chapter:

7.1 Overview of third-party tablet installation

7.1.1 Purpose of this guide

The purpose of this guide is to enable the use of supported third-party tablets with the Earthworks system.

7.1.2 Supported third-party tablets

To run this release of the product on a third-party tablet, use a tablet that meets the following requirements:

- OS: Android 6 to 14
- Supported graphics: OpenGL ES 3.0 or higher
- Display: 8 inches or larger, 16x10 aspect ratio
- Google Chrome
- 2 GB RAM

The recommended tablet is the Samsung Galaxy Tab S7 (SM-T870, SM-T875, or SM-T876B).

7.1.3 Prerequisites

Before starting a third-party tablet installation:

- Make sure that the EC520 controller has the required OS and firmware installed, and that the controller's Wi-Fi access point is enabled. For more information, refer to *Earthworks Configure and Connect Guide*.
- It is strongly recommended that you review this entire guide. This will ensure you understand the full installation workflow.

7.1.4 Equipment list

> **NOTE –**For users who supply their own display, the kit still includes a TD5x0 display harness, which must be used for connecting to the AA510:
>
> - A PC
> - Optionally, a cable to connect the tablet to a cigarette lighter for power

7.2 Third-party tablet installation

7.2.1 Download the app to the tablet

1. Turn on the tablet.
2. For Android 6 or 7 devices, navigate to Settings > Security > Unknown Source, and allow installation of apps from unknown sources.
3. Navigate to Settings > WLAN and connect to the EC520 controller:
   - The name of the access point is the SSID set during the configuration of the controller.
   - The password of the access point is the password set during the configuration of the controller.

   > **NOTE –** Some implementations of the Android operating system do not automatically reconnect to Wi-Fi access points that are not connected to the Internet. If a Network Error message displays after you turn the tablet off, and then turn it on again, check that Wi-Fi is enabled on the tablet, and that the controller is connected.
4. In a web browser, browse to: 192.168.168.1. The Web Interface log-in screen displays.

   > **TIP –** Earthworks supports the Google Chrome browser to run the Web Interface.
5. Log in to the Web Interface. The Web Interface Home screen displays.
6. Navigate to Advanced > Install Operator App. The Install Operator App screen displays.
7. Select Install Direct. The Earthworks firmware download begins.
8. On your device, locate the downloaded *launcher.apk* file, and select it.
9. Follow the instructions on the screen. For Android 8 and above, if the install is blocked by Android, set Allow from this source to ON.
10. If prompted, allow the plugin to access photos, media, and files on your device.

7.2.2 Locate and attach the tablet

In the cab:

1. Decide on how to attach the supported third-party tablet in the cab. We do not suggest locations or methods of attachment.

2. Optionally, attach the cable to connect the display to a cigarette lighter for power.

7.2.3 Validate the install

> **NOTE –** To view the log-in screen, make sure at least one operator is configured, and that maintenance lockout is not active. For more information, refer to the *Configuration Guide*.

1. Turn on the tablet.
2. Locate and tap the app icon. If one of the following displays, then the tablet is connected to the EC520 controller:
   - the log-in screen
   - the *No operator configured for this machine* error message
   - the *Maintenance Lockout Active* error message

It is recommended that you install Android updates when available to keep the system up to date.

8

Torque charts

8.1 Imperial

Maximum torque values in FT-LB and Nm. Applicable to clean, dry, coarse threads in iron.

Nominal size (inches)										
Grade	Torque	1/4	5/16	3/8	7/16	1/2	9/16	5/8	3/4	1”
SAE 2	FT-LB	6	11	19	30	45	66	93	150	300
	Nm	8	15	26	41	61	89	126	203	407
SAE 5	FT-LB	9	18	31	50	75	110	150	250	580
	Nm	12	24	42	68	102	149	203	339	786
SAE 8	FT-LB	13	29	46	75	115	165	225	370	890
	Nm	18	39	62	102	156	224	305	502	1207
SOCKET HEAD	FT-LB	14	30	50	80	120	175	240	390	960
	Nm	19	41	68	108	163	237	325	529	1302

8.2 Metric

Maximum torque values in FT-LB and Nm. Applicable to clean, dry, coarse threads in iron.

Nominal size (mm)									
Grade	Torque	6	8	10	12	14	16	18	20
4.6, 4.8	FT-LB	3	8	16	27	43	60	90	120
	Nm	4	11	22	37	58	81	122	163
8.8	FT-LB	7	17	33	59	100	130	180	280
	Nm	10	23	45	80	136	176	244	380

			Nominal size (mm)							
	Grade	Torque	6	8	10	12	14	16	18	20
10.9	10.9	FT-LB	10	25	48	85	130	200	280	400
		Nm	14	34	65	115	176	271	380	542
12.9	12.9	FT-LB	12	29	57	100	150	230	–	–
		Nm	16	39	77	136	203	312	–	–

Notices

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